Notes for MAT4006 - Introduction to Coding Theory

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Chapter 1

Fundamentals of Coding Theory

1.1 The Language of Codes

Let A be a finite set. We refer to A as a **code alphabet** , and an element of A is called a **code symbol**. If $|A| = q$, the set A is said to be a q-ary alphabet.

Example 1.1. $A = \{0, 1\}$ is a binary alphabet. If $|A| = q$, we often take $A = \{0, 1, \ldots, q-1\}$.

Definition 1.2.

1. An element $x \in A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}}$ is called a **q-ary word of length n** over A.
2. Let $C \subset A^n$ be a non-empty subset of A^n. The set C is called a **q-ary (block) code of length n** over A.
3. An element $c \in C$ is called a **codeword** in C.
4. The number of codewords in C, denoted by $|C|$, is called the **size** of C.
5. The **rate** of C is defined to be $R := \frac{\log_q |C|}{n}$.

A code of length n and size M is called an (n, M) code.

Example 1.3. Let $A = \{0, 1\}$. Define $C_1 \subset A^2, C_2 \subset A^3$ as follows:

(i) $C_1 = \{00, 01, 10, 11\}$ is a $(2, 4)$ code.

(ii) $C_2 = \{000, 011, 101, 110\}$ is a $(3, 4)$ code.

1.2 The Model of Channels

Definition 1.4. A **communication channel** consists of:

- A finite input alphabet A.
- A finite output alphabet B.
- A set of conditional probabilities $\{\Pr(\text{received } b \mid \text{sent } a) : a \in A, b \in B\}$.

A basic channel model is shown below in 1.1

$$a \longrightarrow \boxed{\text{Channel}} \longrightarrow b$$

Figure 1.1: Basic Channel Model

Remark 1.5.

1. $\sum_{b\in B} \Pr(\text{received } b \mid \text{sent } a) = 1$ for all $a \in A$.
2. The model of the channel is known as the **discrete probabilistic channel**. The conditional probabilities can be represented by a stochastic matrix:

$$P = (p_{ij})_{i\in A, j\in B}, \quad p_{ij} = \Pr(j \text{ received} \mid i \text{ sent}).$$

3. We will denote a channel by a triple (A, B, P).

1.3 q-ary Symmetric Channels

Definition 1.6 (Memoryless Channels)**.** A communication channel (A, B, P) is said to be **memoryless** if for $x = (x_1, \ldots, x_n) \in A^n$ and $y = (y_1, \ldots, y_n) \in B^n$, then:

$$\Pr(y \text{ received} \mid x \text{ sent}) = \prod_{i=1}^{n} \Pr(y_i \text{ received} \mid x_i \text{ sent}).$$

Note that this concept is analogous to **independent events** in probability theory.

Definition 1.7. Let $0 \le p \le 1$. A **q-ary symmetric channel** is a memoryless channel (A, A, P) with $|A| = q$ and:

$$\Pr(b \text{ received} \mid a \text{ sent}) = \begin{cases} 1-(q-1)p, & \text{if } a = b \\ p, & \text{if } a \neq b \end{cases}$$

for $a, b \in A$.

Remark 1.8. A q-ary symmetric channel is commonly represented by a diagram. The stochastic matrix is:

$$P = \begin{pmatrix} 1-(q-1)p & p & \cdots & p \\ p & 1-(q-1)p & \cdots & p \\ \vdots & \vdots & \ddots & \vdots \\ p & p & \cdots & 1-(q-1)p \end{pmatrix}$$

Example 1.9 (Binary Symmetric Channel (BSC))**.** Let $q = 2$. The Binary Symmetric Channel has the binary alphabet $A = \{0, 1\}$ and channel probabilities $\Pr(1 \text{ received} \mid 0 \text{ sent}) = \Pr(0 \text{ received} \mid 1 \text{ sent}) = p$.

$$P = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}$$

p is called the **crossover probability** or the **bit-error probability**, and the corresponding graph for the BSC is depicted in Figure 1.2.

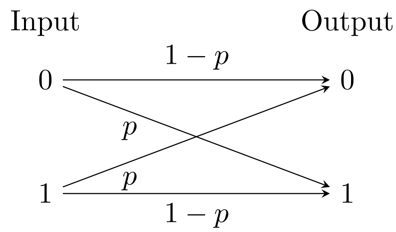


Figure 1.2: Binary Symmetric Channel (BSC)

Example 1.10. Let $C = \{000, 111\}$ be a binary code over $\{0, 1\}$. Consider a Binary Symmetric Channel (BSC) with $p = 0.05$.

Assume a uniform distribution on C, i.e., $\Pr(000) = \Pr(111) = \frac{1}{2}$. Suppose 110 is received. Which of the codewords is more likely to be the one sent?

Compute: $\Pr(111 \text{ sent} \mid 110 \text{ received}) = \frac{\Pr(110 \text{ received}|111 \text{ sent}) \cdot \Pr(111 \text{ sent})}{\Pr(110 \text{ received})}$

Since the channel is **memoryless**:

$$\begin{aligned}\Pr(110 \text{ received} \mid 111 \text{ sent}) &= \Pr(1 \text{ rec} \mid 1 \text{ sent})^2 \cdot \Pr(0 \text{ rec} \mid 1 \text{ sent}) \\ &= (0.95)^2 \cdot 0.05\end{aligned}$$

Note that q-ary channels are defined to be memoryless.

Similarly:

$$\Pr(110 \text{ received} \mid 000 \text{ sent}) = \frac{\Pr(110 \text{ rec} \mid 000 \text{ sent}) \cdot \Pr(000 \text{ sent})}{\Pr(110 \text{ received})}$$

$$\Pr(110 \text{ received} \mid 000 \text{ sent}) = (0.05)^2 \cdot 0.95$$

Since $(0.95)^2 \cdot 0.05 > (0.05)^2 \cdot 0.95$, we have:

$$\Pr(110 \text{ received} \mid 111 \text{ sent}) > \Pr(110 \text{ received} \mid 000 \text{ sent})$$

Note: We compare $\Pr(110 \text{ rec} \mid 000 \text{ sent})$ and $\Pr(110 \text{ rec} \mid 111 \text{ sent})$. This is the **Maximum Likelihood (ML) decoding rule**.

1.4 Decoding Rules and Hamming Distance

Definition 1.11. Let C be an (n, M) code over an alphabet A and let $W = (A, A, P)$ be a memoryless channel. A **decoder** for C with respect to W is a function $\mathcal{D} : A^n \to C$. The **average decoding error probability** P_e of $\mathcal{D}$ is defined by:

$$P_e = \sum_{c \in C} P_e(c) Pr(c),$$

where

$$P_e(c) := \sum_{y:\mathcal{D}(y)\neq c} Pr(y \text{ received} \mid c \text{ sent}).$$

Definition 1.12 (Maximum Likelihood Decoding (MLD))**.** Suppose a code C is used to transmit information over a channel. If a word y is received, the MLD rule will decode y to x if:

$$\Pr(y \text{ received} \mid x \text{ sent}) = \max_{x'\in C} \Pr(y \text{ received} \mid x' \text{ sent}).$$

(Ties are broken arbitrarily).

Remark 1.13. Note the difference:

- Posterior probability: $\Pr(x \text{ sent} \mid y \text{ received})$.
- Likelihood: $\Pr(y \text{ received} \mid x \text{ sent})$.
- Prior probability: $\Pr(x \text{ sent})$.

Example 1.14. Suppose $C \subset A^n$ is used over BSC with crossover probability $p < \frac{1}{2}$. Then

$$\Pr(y \text{ received} \mid x \text{ sent}) \xrightarrow{\text{memoryless}} \prod_{i=1}^{n} \Pr(y_i \text{ received} \mid x_i \text{ sent}) = p^e(1-p)^{n-e} =: p^{d(x,y)}(1-p)^{n-d(x,y)}.$$

where e is the number of places where y and x differ.

Since $p < \frac{1}{2}$, we have $1 - p > \frac{1}{2}$. Smaller values of e implies larger value of $p^e(1-p)^{n-e}$. Therefore,

$$\Pr(y \text{ received} \mid x \text{ sent}) = p^e(1-p)^{n-e}$$

is maximized by a codeword $x \in C$ for which e is as small as possible.

Definition 1.15. Let $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in A^n$. The **Hamming distance** from x to y, denoted by $d(x, y)$, is defined to be:

$$d(x,y) = \sum_{i=1}^{n} \delta(x_i, y_i), \quad \text{where } \delta(u,v) = \begin{cases} 0 & \text{if } u = v \\ 1 & \text{if } u \neq v \end{cases}$$

Example 1.16. (i) Let $A^5 = \{0,1\}^5$ and $x, y, z \in A^5$.

$x = 01010, \quad y = 01101, \quad z = 11101.$ and $d(x,y) = 3, \quad d(x,z) = 4, \quad d(y,z) = 1.$

(ii) Let $x, y \in \{0, 1, \ldots, 4\}^4$.

$x = 1234, \quad y = 1423.$ and $d(x,y) = 3.$

Proposition 1.17. *The Hamming distance is a metric. Let $x, y, z \in A^n$. Then:*

1. $0 \le d(x,y) \le n$ *(Positivity).*
2. $d(x,y) = 0 \iff x = y$.

3. $d(x, y) = d(y, x)$ *(Symmetry).*
4. $d(x, y) + d(y, z) \geq d(x, z)$ *(Triangle Inequality).*

Proof of (iv). $d(x, z) = \sum_{i=1}^{n} \delta(x_i, z_i)$, where $\delta(x_i, z_i) = \begin{cases} 0 & \text{if } x_i = z_i \\ 1 & \text{if } x_i \neq z_i \end{cases}$.

We want to show $\delta(x_i, y_i) + \delta(y_i, z_i) \geq \delta(x_i, z_i)$.

- If $x_i = z_i$, RHS $= 0$, easily we see the inequality holds.
- If $x_i \neq z_i$, RHS $= 1$. Suppose LHS $< 1 \implies$ LHS $= 0$, then we have $x_i = y_i = z_i$, which contradicts to our assumption.

Summing over all i proves the inequality. □

1.5 Minimum Distance Decoding

Definition 1.18. Suppose a code C is used over a channel. If y is received, the **minimum distance decoding rule** (nearest neighbor decoding) will decode y to $x \in C$ if:

$$d(x, y) = \min_{c \in C} d(c, y).$$

Theorem 1.19. *For a q-ary symmetric channel with parameter $p < 1/q$, the MLD rule is the same as the minimum distance decoding rule.*

Proof. Suppose $C \subset A^n$ is used over a q-ary symmetric channel. Let $e = d(x, y)$ be the number of places where y and x differ.

$$\Pr(y \text{ received} \mid x \text{ sent}) = p^e (1 - (q-1)p)^{n-e} = p^{d(x,y)} (1 - (q-1)p)^{n-d(x,y)}.$$

Since $p < 1/q$, we have $p < 1 - (q-1)p$. Smaller values of e (distance) imply larger values of the probability. Thus, maximizing the likelihood is equivalent to minimizing the distance $d(x, y)$. □

Remark 1.20.
1. If $p > 1/q$, the MLD rule is the "Maximum Distance Rule", opposite to the above conclusion.
2. MLD Rule $\iff$ Find x minimizing $d(x, y)$ $\iff$ Minimum Distance Rule

1.6 Distance of a Code and Error Detection

Definition 1.21. For a code C with $|C| \geq 2$, the **(minimum) distance** of C, denoted by $d(C)$, is:

$$d(C) = \min\{d(x, y) \mid x, y \in C, x \neq y\}.$$

Remark 1.22. A code of length n, size M, and distance d is referred to as an (n, M, d) code. The numbers n, m, d are called parameters of the code.

Definition 1.23 (Error Detection)**.** Let u be a positive integer. We say a code $C \subset A^n$ can **detect** u errors if for any $x \in C$ and $y \in A^n$ such that $1 \leq d(x,y) \leq u$, we have $y \notin C$.

Theorem 1.24. *A code $C \subset A^n$ can detect u errors if and only if $d(C) \geq u+1$.*

Proof. **(⇒)** Assume C can detect u errors. For any distinct $x, y \in C$, the distance must satisfy $d(x,y) > u$ (otherwise y would be a valid codeword within error range).

$$\implies d(x,y) \geq u+1.$$

By definition, $d(C) = \min_{x \neq y} d(x,y) \geq u+1$.

(⇐) Assume $d(C) \geq u+1$. Take any $x \in C$ and receive y such that $1 \leq d(x,y) \leq u$. Suppose $y \in C$. Then $y \neq x$, so:

$$d(x,y) \geq d(C) \geq u+1.$$

This contradicts $d(x,y) \leq u$. $\implies y \notin C$. Hence, the error is detected. □

1.7 Error Correction and Minimum Distance

Definition 1.25 (Error Correction)**.** Let ν be a positive integer. We say a code $C \subset A^n$ can correct ν errors if there is a decoder

$$\mathcal{D} : A^n \to C$$

such that for any $x \in C$ and $y \in A^n$ with $d(x,y) \leq \nu$, one has

$$\mathcal{D}(y) = x.$$

Theorem 1.26. *A code C can correct ν errors if $d(C) \geq 2\nu + 1$.*

Proof. Let $x \in C$ and y be the received words such that $d(x,y) \leq \nu$. For any $x' \in C$ with $x' \neq x$, we have

$$2\nu + 1 \leq d(C) \leq d(x,x') \leq d(x,y) + d(y,x')$$

by the triangle inequality. Since $d(x,y) \leq \nu$, we have:

$$2\nu + 1 \leq \nu + d(y,x') \implies d(y,x') \geq \nu + 1.$$

Therefore,

$$d(y,x') \geq \nu + 1, \quad \text{for any } x' \in C, x' \neq x.$$

So x is the *only* codeword closest to y. Using the minimum distance decoder, we can decode y to x. □

Theorem 1.27. *If a code $C \subset A^n$ can correct ν errors using the minimum distance decoder, then*

$$d(C) \geq 2\nu + 1.$$

Proof. Suppose $d(C) \le 2\nu$. Let us construct a word $y \in A^n$ that confuses the decoder.

Let $x, x' \in C$ be distinct codewords such that $d(x, x') = d(C)$. Let $I = \{i \mid x_i \ne x'_i\}$ be the set of indices where they differ. So $|I| = d(C) \le 2\nu$.

Fix a subset $J \subset I$ with size $|J| = \lfloor \frac{d(C)}{2} \rfloor$. Set y as follows:

- $y_i = x_i$ for $i \in J$ (Matches x).
- $y_i = x'_i$ for $i \in I \setminus J$ (Matches x').
- $y_i = x_i = x'_i$ for $i \notin I$.

Now we verify the distances (Scaling):

1. Distance to x: y differs from x only in $I \setminus J$.

$$d(x, y) = |I \setminus J| = d(C) - \left\lfloor \frac{d(C)}{2} \right\rfloor = \left\lceil \frac{d(C)}{2} \right\rceil \le \left\lceil \frac{2\nu}{2} \right\rceil = \nu.$$

2. Distance to x': y differs from x' only in J.

$$d(x', y) = |J| = \left\lfloor \frac{d(C)}{2} \right\rfloor \le \left\lfloor \frac{2\nu}{2} \right\rfloor = \nu.$$

Since y is within distance ν of both x and x', the decoder cannot uniquely recover the sent message (it is confused). This contradicts the assumption that C can correct ν errors.

Therefore, we must have $d(C) \ge 2\nu + 1$.

The following diagram illustrates a concrete example to help clarify the separation.

$$\begin{array}{l}
x = (\overbrace{\underbrace{x_1}_{J}, x_2, x_3}^{d(C)}, \ldots, \overbrace{x_{n-1}, x_n}^{n-d(C)}) \\
x' = (x'_1, \underbrace{x'_2, x'_3}_{I\setminus J}, \ldots, x'_{n-1}, x'_n) \\
y = (\underline{y_1}, \underline{y_2, y_3}, \ldots, y_{n-1}, y_n)
\end{array}$$

$\square$

Remark 1.28. For non-probabilistic channels (also known as adversarial / deterministic / combinatorial channels) that can tamper at most ν symbols, we can use a code with distance at least $2\nu + 1$ to protect information transmission.

Chapter 2

Abstract algebra

Consider $C \subset A^n$ where A is an arbitrary finite set. To have more structure (like linearity), we need algebraic operations on A. For example, if $a_1, a_2 \in A$, we want $a_1 + a_2 \in A$. This leads us to Groups, Rings, and Fields.

2.1 Groups

Definition 2.1. A **group** is a nonempty set G with a binary operation "$\cdot$" on G that satisfies the following axioms:

1. **Closure**: $a \cdot b \in G$ for every $a, b \in G$. (Note: usually implied by the definition of binary operation).
2. **Associativity**: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for every $a, b, c \in G$.
3. **Identity element**: There exists an element in G, denoted by 1, such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for every } a \in G.$$
4. **Inverse element**: For every $a \in G$, there exists an element in G, denoted by a^{-1}, such that
$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

Definition 2.2 (Abelian Group)**.** A group is called **commutative** or **abelian** if $a \cdot b = b \cdot a$ for every $a, b \in G$.

Notation of powers: For an element $a \in G$ and a positive integer n, the notation a^n stands for
$$\underbrace{a \cdot a \cdot \cdots \cdot a}_{n \text{ times}}.$$
Moreover, a^0 denotes the identity element $1 \in G$.

Notational Conventions: There are two main notational conventions for groups: Multiplicative (usual) and Additive.

	Operation	**Identity**	**Power**	**Inverse**
Additive group	$+$	0	$n \cdot a$ (or na)	$-a$
Multiplicative group	$\cdot$	1	a^n	a^{-1}

Remark:

- In additive notation, $a - b$ stands for $a + (-b)$.
- In multiplicative notation, ab stands for $a \cdot b$.

Example 2.3. $(\mathbb{Z}, +)$ is an additive group.

Example 2.4. Let $n\mathbb{Z} = \{nx \mid x \in \mathbb{Z}\}$ (also denoted by (n)). For instance,

$$(2) = 2\mathbb{Z} = \{\dots, -2, 0, 2, \dots\}.$$

Then $(n\mathbb{Z}, +)$ is an additive group.

2.2 Rings

Definition 2.5 (Ring)**.** A ring is a nonempty set R with two binary operations "$\cdot$" and "$+$" on R that satisfy:

1. R is an abelian group with respect to (w.r.t.) "$+$".
2. Closure w.r.t. "$\cdot$" ($\cdot : R \times R \to R$).
3. Associativity w.r.t. "$\cdot$":

$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for every } a, b, c \in R.$$

4. Distributivity:

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (b + c) \cdot a = b \cdot a + c \cdot a$$

for every $a, b, c \in R$.

Remark 2.6 (Convention)*.* When writing expressions, "$\cdot$" takes precedence over "$+$" so:

$$c + a \cdot b = c + (a \cdot b).$$

Definition 2.7 (Zero Element)**.** The identity element of "$+$" is called the *zero element* in R.

Definition 2.8 (Ring with Identity)**.** A ring with the identity is a ring R in which "$\cdot$" has an identity: There exists an element in R, denoted by 1, such that

$$1 \cdot a = a \cdot 1 \quad \text{for any } a \in R.$$

Definition 2.9 (Commutative Ring)**.** A commutative ring is a ring R in which "$\cdot$" is commutative:

$$a \cdot b = b \cdot a \quad \text{for every } a, b \in R.$$

2.3 Fields

Definition 2.10. A field F is a commutative ring in which the nonzero elements form a group w.r.t. "$\cdot$", i.e.,

$$F^* := F \setminus \{0\} \quad \text{is a group w.r.t. "}\cdot\text{".}$$

(This implies $(F^*, \cdot)$ has the identity element and inverse w.r.t. "$\cdot$").

Lemma 2.11. *Let $a, b \in F$ where F is a field. Then:*

1. $0 \cdot a = 0$ *for any* $a \in F$.
2. $(-1) \cdot a = -a$.
3. $ab = 0$ *implies* $a = 0$ *or* $b = 0$.

Proof. 1. Consider the expression $0 \cdot a$:

$$0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a.$$

By adding the inverse element of $0 \cdot a$ in both sides of the equation, we get $0 = 0 \cdot a$.

2. We compute the sum $(-1) \cdot a + a$:

$$(-1) \cdot a + a = (-1) \cdot a + 1 \cdot a = (-1 + 1) \cdot a = 0 \cdot a.$$

Using the result from (1), we have $0 \cdot a = 0$. Thus, $(-1) \cdot a$ is the additive inverse of a, i.e., $(-1) \cdot a = -a$.

3. Suppose $ab = 0$ and $a \neq 0$. Since F is a field, a^{-1} exists. Multiplying by a^{-1} from the left:

$$a^{-1}(ab) = a^{-1} \cdot 0.$$

Using associativity and the result from (1) (where $a^{-1} \cdot 0 = 0$):

$$(a^{-1}a)b = 0 \implies 1 \cdot b = 0 \implies b = 0.$$

The same logic holds if $b \neq 0$, which implies $a = 0$.

□

Example 2.12. 1. $(\mathbb{Z}, +, \cdot)$ is a ring, where the operations are the standard addition and multiplication. This is called the **ring of integers**.

2. $\mathbb{R}, \mathbb{C}, \mathbb{Q}$ are fields.

Example 2.13 (A Finite Field)**.** Consider the set $(\{0,1\},+,\cdot)$ with operations defined by the following tables:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

This is a field.

Remark 2.14. A field containing a finite number of elements is called a *finite field*.

2.4 The Ring $\mathbb{Z}_m$

Definition 2.15 (Congruence)**.** Let a, b, and $m > 1$ be integers. We say a is congruent to b modulo m, written as

$$a \equiv b \pmod m$$

If $m \mid (a-b)$, i.e., m divides $a-b$.

Remark 2.16. 1. Given a and $m > 1$, by the long division, we have

$$a = qm + b$$

where $0 \leq b \leq m-1$ and b is <u>uniquely determined</u> by a and m. (This implies any integer a is congruent to exactly one of $0, \ldots, m-1$ modulo m).

2. The integer b is called the <u>remainder</u> of a divided by m, denoted by $a \pmod m$. (So a is congruent to the remainder of a).

3. Properties of the modulo operation:

$$\left.\begin{aligned} a \equiv b \pmod m \\ c \equiv d \pmod m \end{aligned}\right\} \implies \begin{aligned} a \pm c \equiv b \pm d \pmod m \\ a \cdot c \equiv b \cdot d \pmod m \end{aligned}$$

For $m > 1$, we denote the set $\{0, \ldots, m-1\}$ by $\mathbb{Z}_m$ and define "+" and "$\cdot$" in $\mathbb{Z}_m$ by:

$$\begin{aligned} a + b &= (a + b \pmod m) \\ a \cdot b &= (a \cdot b \pmod m) \end{aligned}$$

(where the sum and product inside the parentheses are standard addition and multiplication).

Proposition 2.17. *One can show $(\mathbb{Z}_m, +, \cdot)$ is a ring.*

Proof. Let $\mathbb{Z}_m = \{0, 1, \ldots, m-1\}$. The operations are defined by $a + b = (a+b)_{\mathbb{Z}} \pmod m$ and $a \cdot b = (a \cdot b)_{\mathbb{Z}} \pmod m$.

We verify the ring axioms step by step.

1. $(\mathbb{Z}_m, +)$ **is an Abelian Group**

- **Closure:** By definition of the modulo operation, for any $a, b \in \mathbb{Z}_m$, the remainder is in $\mathbb{Z}_m$.
- **Associativity:** For any $a, b, c \in \mathbb{Z}_m$:

$$(a+b)+c \equiv (a+b)_{\mathbb{Z}} + c \equiv a + (b+c)_{\mathbb{Z}} \equiv a + (b+c) \pmod{m}.$$

  This holds because integer addition is associative.
- **Identity Element:** The element $0 \in \mathbb{Z}_m$ acts as the identity:

$$a + 0 \equiv a \pmod{m} \implies a + 0 = a.$$

- **Inverse Element:** For any $a \in \mathbb{Z}_m$:
  - If $a = 0$, the inverse is 0 (since $0 + 0 = 0$).
  - If $a \neq 0$, the additive inverse is $m - a$, because:

$$a + (m - a) = m \equiv 0 \pmod{m}.$$

  Since $0 \leq a < m$, we have $0 < m - a < m$, so the inverse is in $\mathbb{Z}_m$.
- **Commutativity:** Since integer addition is commutative ($a + b = b + a$), it follows that:

$$a + b \equiv b + a \pmod{m}.$$

2. $(\mathbb{Z}_m, \cdot)$ **is Associative**

For any $a, b, c \in \mathbb{Z}_m$, we rely on the associativity of integer multiplication:

$$(a \cdot b) \cdot c \equiv (ab)_{\mathbb{Z}} \cdot c \equiv a \cdot (bc)_{\mathbb{Z}} \equiv a \cdot (b \cdot c) \pmod{m}.$$

3. Distributivity

The distributive laws hold because they hold in $\mathbb{Z}$. For any $a, b, c \in \mathbb{Z}_m$:

$$a \cdot (b+c) \equiv a \cdot (b+c)_{\mathbb{Z}} \equiv (ab+ac)_{\mathbb{Z}} \equiv a \cdot b + a \cdot c \pmod{m}.$$

Similarly, $(a + b) \cdot c = a \cdot c + b \cdot c$.

$\square$

Example 2.18. Consider $\mathbb{Z}_4 = \{0, 1, 2, 3\}$. The operation tables are given by:

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$\cdot$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

Question: Is $(\mathbb{Z}_4 \setminus \{0\}, \cdot)$ a group?
Answer: No. Observe the row for 2 in the multiplication table:

$$2 \cdot 0 = 0, \quad 2 \cdot 1 = 2, \quad 2 \cdot 2 = 0, \quad 2 \cdot 3 = 2.$$

The value 1 (the identity) never appears. Thus, 2 does not have an inverse with respect to "$\cdot$".

Q: When is $\mathbb{Z}_m$ a field?

Theorem 2.19. *$\mathbb{Z}_m$ is a field if and only if m is a prime.*

Proof. ($\implies$) Suppose m is composite. Let $m = ab$ where integers a, b satisfy $1 < a, b < m$. In the ring $\mathbb{Z}_m$, we have:

$$0 \equiv m \pmod{m} \implies 0 \equiv ab \pmod{m}.$$

Thus, $ab = 0$ in $\mathbb{Z}_m$. However, $a \neq 0$ and $b \neq 0$ in $\mathbb{Z}_m$. This implies that $\mathbb{Z}_m$ has zero divisors. But in any field, $ab = 0$ implies $a = 0$ or $b = 0$. This contradiction shows that if m is composite, $\mathbb{Z}_m$ is not a field.

($\impliedby$) Suppose m is prime. We need to show that $\mathbb{Z}_m^* = \mathbb{Z}_m \setminus \{0\}$ is an abelian group with respect to multiplication. Since we already know $(\mathbb{Z}_m, +, \cdot)$ is a commutative ring, it suffices to show the existence of a multiplicative inverse for each $a \in \mathbb{Z}_m^*$.

Let $a \in \mathbb{Z}_m$ with $a \neq 0$. Since m is prime and $1 \leq a < m$, the greatest common divisor is $\gcd(a, m) = 1$. By *Bézout's Identity*, there exist integers u, v such that:

$$\begin{aligned} ua + vm &= 1 \\ ua &= 1 - vm \\ ua &\equiv 1 \pmod{m} \\ u &= a^{-1} \text{ in } \mathbb{Z}_m \end{aligned}$$

Since every non-zero element has an inverse, $\mathbb{Z}_m$ is a field. □

2.5 Ring of Polynomimals

Definition 2.20 (Ring of Polynomials)**.** Let F be a field. Define the following set

$$F[x] := \left\{ \sum_{i=0}^{n} a_i x^i \mid a_i \in F \text{ for all } i; n \geq 0 \right\}$$

This set is called the ring of polynomials over F. (Note: x is called the *indeterminate* and a_i is called the *coefficient*).

Check: the set of polynomials form a ring.

An element $f(x) \in F[x]$ is called a polynomial over F. We write $f(x) = \sum_{i=0}^{n} a_i x^i$.

Definition 2.21 (Degree)**.** If the leading coefficient $a_n \neq 0$, the integer n is called the degree of f, denoted by $\deg(f)$. More precisely,

$$\deg(f) := \max_{0 \leq i \leq n : a_i \neq 0} i$$

We define $\deg(0) = -\infty$.

Definition 2.22 (Monic Polynomial)**.** A nonzero polynomial f of degree n is said to be monic if $a_n = 1$.

Definition 2.23 (Reducible Polynomial)**.** A polynomial $f \in F[x]$ with $\deg(f) > 0$ is said to be reducible over F if there exist two polynomials $g, h \in F[x]$ such that

$$\deg(f) > \deg(g), \quad \deg(f) > \deg(h) \quad \text{and} \quad f(x) = g(x)h(x).$$

Otherwise, f is said to be irreducible over F.

Example 2.24.

1. **A reducible polynomial in $\mathbb{Z}_3[x]$:**
   Let $f(x) = x^4 + 2x^6 \in \mathbb{Z}_3[x]$.
   - $\deg(f) = 6$, f is not monic.
   - $f(x) = x^4(1 + 2x^2)$ is reducible.
2. **A polynomial in $\mathbb{Z}_2[x]$:**
   Let $g(x) = 1 + x + x^2 \in \mathbb{Z}_2[x]$.
   - $\deg(g) = 2$, g is monic.
   - *Evaluation (Substitute x by 0 and 1):* Since $\mathbb{Z}_2 = \{0, 1\}$, we calculate:

     $$g(0) = 1, \quad g(1) = 1 + 1 + 1 = 3 = 1.$$

     Let $b \in \mathbb{Z}_2$. Then $g(b) = 1 + b + b^2 \in \mathbb{Z}_2$.
     Since $g(x)$ has no roots in $\mathbb{Z}_2$, $g(x)$ has no linear factors. $\implies$ g is irreducible.
   - *Note on Degree-1 polynomials over $\mathbb{Z}_2$*: The only degree-1 polynomials are x and $x + 1$.

Remark 2.25. If $f(x) = \sum_{i=0}^{n} a_i x^i \in F[x]$ and $\beta \in F$, then

$$f(\beta) = a_0 + a_1\beta + \cdots + a_n\beta^n$$

yields an element in F.

Definition 2.26 (Subfields)**.** Let E, F be two fields and $F \subseteq E$. The field F is called a subfield of E if the addition and multiplication of E, when restricted to F, are the same as those of F. E is also called an extension field of F.

Example 2.27.
- $\mathbb{Q}$ is a subfield of $\mathbb{R}$.
- $\mathbb{C}$ is an extension field of $\mathbb{R}$.

Definition 2.28 (Root)**.** Let F be a field and E be an extension field of F. An element $\beta \in E$ is a root of a polynomial $f(x) \in F[x]$ if the equality $f(\beta) = 0$ holds in E.

Example 2.29. Consider $f(x) = 1 + x^2 \in \mathbb{R}[x]$ (where $F = \mathbb{R}$). Note that f is irreducible over $\mathbb{R}$.

Let β be the imaginary unit in $\mathbb{C}$ (where $E = \mathbb{C}$). We have:

$$f(\beta) = 1 + \beta^2 = 0 \quad \text{in } \mathbb{C}.$$

So β is a root of $f(x)$ in $\mathbb{C}$.

We can define congruence on $F[x]$.

Proposition 2.30 (Division Algorithm for Polynomials)**.** *Let $f(x) \in F[x]$, with $\deg(f) = n \geq 1$. Then for any $g(x) \in F[x]$, there exists a unique pair $(s(x), r(x))$ of polynomials over F with $\deg(r) < \deg(f)$ such that*

$$g(x) = s(x)f(x) + r(x).$$

Definition 2.31 (Remainder)**.** The polynomial $r(x)$ is called the remainder of $g(x)$ divided by $f(x)$, denoted by $g(x) \pmod{f(x)}$.

Example 2.32. Let $f(x) = 1 + x^2 \in \mathbb{Z}_5[x]$ and $g(x) = x + 2x^4 \in \mathbb{Z}_5[x]$.

Computation:

$$
\begin{array}{r}
2x^2 - 2 \\
x^2 + 1 \overline{) \, 2x^4 \qquad\quad + x} \\
\underline{2x^4 + 2x^2 \quad} \\
-2x^2 + x \\
\underline{-2x^2 \qquad - 2} \\
x + 2
\end{array}
$$

Thus, we have:

$$g(x) = (2x^2 - 2)f(x) + (x + 2)$$

In $\mathbb{Z}_5$, since $-2 \equiv 3$, the quotient becomes $2x^2 + 3$.
So:

$$g(x) \equiv x + 2 \pmod{f(x)}$$

Definition 2.33. For a given polynomial $f(x) \in F[x]$ with $\deg(f) = n \geq 1$, we define the set of remainders as:

$$F[x]/(f(x)) := \left\{ \sum_{i=0}^{n-1} a_i x^i \mid a_i \in F \right\}$$

(This is analogous to $\mathbb{Z}_m = \{0, 1, \ldots, m-1\}$).

Integers ($\mathbb{Z}$)	**Polynomials** ($F[x]$)
Integer m	Polynomial $f(x)$
Prime Number	Irreducible Polynomial
$\mathbb{Z}_m$ (Quotient Ring)	$F[x]/(f(x))$ (Quotient Ring)
Operations (Arithmetic)	
$a+b=(a+b) \pmod m$ $ab=(ab) \pmod m$	$g(x)+h(x) \equiv (g(x)+h(x)) \pmod{f(x)}$ $g(x)h(x) \equiv g(x)h(x) \pmod{f(x)}$
Algebraic Structure Properties	
$\mathbb{Z}_m$ is always a commutative ring.	$F[x]/(f(x))$ is always a commutative ring.
Theorem: $\mathbb{Z}_m$ is a field $\iff$ m is a prime.	**Theorem:** $F[x]/(f(x))$ is a field $\iff$ $f(x)$ is irreducible.

Theorem 2.34. *Let $f(x)$ be a polynomial over a field F with $\deg(f) = n \geq 1$. Then $F[x]/(f(x))$ together with "+" and "·" defined in the table (using congruence) forms a ring.*

Furthermore, $F[x]/(f(x))$ is a field if and only if $f(x)$ is irreducible over F.

Proof. The proof is exactly the same as we prove $\mathbb{Z}_m$ is a field. We omit the details here. □

Example 2.35.

1. $\mathbb{R}$ is a field. The polynomial $1+x^2$ is irreducible over $\mathbb{R}$. Therefore, the quotient ring

$$\mathbb{R}[x]/(1+x^2) \quad \text{is a field.}$$

This set can be written as:

$$\{a_0 + a_1 x \mid a_0, a_1 \in \mathbb{R}\} \cong \mathbb{C}$$

2. Consider $1+x^2$ over $\mathbb{Z}_2$. Question: Is $\mathbb{Z}_2[x]/(1+x^2)$ a field?

We need to check whether $1+x^2$ is irreducible over $\mathbb{Z}_2$. Let $f(x) := 1+x^2$. Evaluate at $x=1$:

$$f(1) = 1+1 = 0$$

Since 1 is a root of $f(x)$ in $\mathbb{Z}_2$, the polynomial can be factored:

$$\implies f(x) = (1+x)^2$$

Since $f(x)$ is reducible,

$$\implies \mathbb{Z}_2[x]/(f(x)) \quad \text{is \underline{not} a field.}$$

3. **Question:** What about the quotient ring $\mathbb{Z}_2[x]/(1+x+x^2)$?

   Let $g(x) := 1 + x + x^2$. We check for roots in $\mathbb{Z}_2$:

$$\begin{aligned} g(0) &= 1 \\ g(1) &= 1+1+1 = 3 = 1 \quad (\text{in } \mathbb{Z}_2) \end{aligned}$$

   Since $g(x)$ has no roots in $\mathbb{Z}_2$:

$$\implies g(x) \text{ has no linear factors in } \mathbb{Z}_2[x].$$

$$\implies g(x) \text{ is irreducible over } \mathbb{Z}_2.$$

   Therefore, the quotient forms a field. The elements are all polynomials of degree less than 2 with coefficients in $\mathbb{Z}_2$:

$$\mathbb{Z}_2[x]/(1+x+x^2) = \{0, 1, x, 1+x\}.$$

Question: How many elements are in $F[x]/(f(x))$?

By the Division Algorithm, every polynomial can be uniquely represented by a remainder of degree strictly less than n.

$$F[x]/(f(x)) = \left\{ \sum_{i=0}^{n-1} g_i x^i \;\middle|\; g_i \in F \right\}$$

Writing this out explicitly, an arbitrary element looks like:

$$g_0 + g_1 x + g_2 x^2 + \cdots + g_{n-1} x^{n-1}$$

where each coefficient g_i belongs to the base field F.

- Choice for g_0: $|F|$ possibilities ($\because g_0 \in F$).
- Choice for g_1: $|F|$ possibilities ($\because g_1 \in F$).
- ...
- Choice for g_{n-1}: $|F|$ possibilities ($\because g_{n-1} \in F$).

Since there are n independent coefficients to choose:

$$|F[x]/(f(x))| = \underbrace{|F| \cdot |F| \cdots |F|}_{n \text{ times}} = |F|^n$$

Definition 2.36 (Characteristic)**.** Let F be a field. The <u>characteristic</u> of F is the <u>least positive integer</u> p such that

$$p \cdot 1 = 0$$

where 1 is the multiplicative identity of F. If no such integer exists, the characteristic is defined to be zero.

Denote the characteristic of F by $\text{char}(F)$.

Note: The notation $p \cdot 1$ represents the p-th (additive) power of 1:

$$p \cdot 1 = \underbrace{1 + 1 + \cdots + 1}_{p \text{ times}}$$

Example 2.37.

$$\text{char}(\mathbb{R}) = 0$$

$$\text{char}(\mathbb{C}) = 0$$

Theorem 2.38. *The characteristic of a field is either* 0 *or a* prime.

Proof. We will show the characteristic cannot be 1 or any other composite number.

It is clear $1 \cdot 1 \neq 0$. So 1 cannot be the characteristic of any field.

Suppose the characteristic of a field F is $p = n \cdot m$ where $1 < n, m < p$. Define $a, b \in F$ by $a = n \cdot 1$ and $b = m \cdot 1$. Then in F, we have:

$$a \cdot b = \left(\sum_{i=1}^{n} 1\right)\left(\sum_{j=1}^{m} 1\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} 1 = (nm) \cdot 1 = p \cdot 1 = 0$$

$$\implies a \cdot b = 0 \quad \& \quad a \cdot b \in \mathbb{F}$$

Since F is a field, it has no zero divisors, so:

$$\implies a = 0 \quad \text{or} \quad b = 0.$$

$$\implies n \cdot 1 = 0 \quad \text{or} \quad m \cdot 1 = 0.$$

This contradicts the assumption that p is the characteristic (which implies p is the **least** positive integer such that $p \cdot 1 = 0$), because $n < p$ and $m < p$.

Therefore, p cannot be composite. $\square$

Remark 2.39 (Finite Fields Setup). **Question:** What sizes of a finite field can be?

Let F be a finite field with $\text{char}(F) = p$. Then F contains the set generated by 1:

$$F \supset \{0, 1, 2 \cdot 1, \ldots, (p-1) \cdot 1\}$$

This subset forms a subfield isomorphic to $\mathbb{Z}_p$:

$$\mathbb{Z}_p = \{0, 1, 2 \cdot 1, \ldots, (p-1) \cdot 1\}$$

i.e., $\mathbb{Z}_p$ can be viewed as a subfield of F.

Theorem 2.40. *A finite field F with* $\text{char}(F) = p$ *contains p^n elements for some integer $n \geq 1$.*

Proof. Choose $\alpha_1 \in F^* = F \setminus \{0\}$. We claim that the elements $0 \cdot \alpha_1, 1 \cdot \alpha_1, \ldots, (p-1) \cdot \alpha_1$ are all distinct.

Indeed, if $i \cdot \alpha_1 = j \cdot \alpha_1$ for $0 \leq i \leq j \leq p-1$, then:

$$(j-i) \cdot \alpha_1 = 0.$$

Since $\alpha_1 \in F^*$ (so $\alpha_1 \neq 0$) and F has no zero divisors, we must have $(j-i) \cdot 1 = 0$. Given that $\mathrm{char}(F) = p$ and $0 \leq j-i \leq p-1$, this implies $j-i=0$, so $i=j$. Thus, these p elements are distinct.

If $F = \{0 \cdot \alpha_1, \ldots, (p-1) \cdot \alpha_1\}$, then $|F| = p$ and we are done.

Otherwise, choose $\alpha_2 \in F \setminus \{0 \cdot \alpha_1, \ldots, (p-1) \cdot \alpha_1\}$. We claim that the linear combinations $a_1\alpha_1 + a_2\alpha_2$ are distinct for all $0 \leq a_1, a_2 \leq p-1$. Suppose:

$$a_1\alpha_1 + a_2\alpha_2 = b_1\alpha_1 + b_2\alpha_2 \quad \text{for } 0 \leq a_i, b_i \leq p-1.$$

Rearranging terms:

$$(a_1 - b_1)\alpha_1 = (b_2 - a_2)\alpha_2.$$

Suppose $b_2 - a_2 \neq 0$. Then $(b_2 - a_2)$ is invertible in $\mathbb{Z}_p$, and we can write:

$$\alpha_2 = (b_2 - a_2)^{-1}(a_1 - b_1)\alpha_1.$$

Since coefficients are in $\mathbb{Z}_p$, this implies α_2 is a multiple of α_1 (i.e., $\alpha_2 \in \mathrm{span}\{\alpha_1\}$), which contradicts the choice of α_2. Thus, we must have $b_2 = a_2$, which in turn implies $a_1 = b_1$. Therefore, the set $\{a_1\alpha_1 + a_2\alpha_2 \mid a_i \in \mathbb{Z}_p\}$ contains $p \times p = p^2$ distinct elements.

As $|F| < \infty$, we can continue this way to obtain $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n$ such that:

$$\alpha_i \in F \setminus \{a_1\alpha_1 + \cdots + a_{i-1}\alpha_{i-1} \mid a_k \in \mathbb{Z}_p\}$$

for all $2 \leq i \leq n$.

Eventually, the process terminates, and we have:

$$F = \{a_1\alpha_1 + \cdots + a_n\alpha_n \mid a_1, \ldots, a_n \in \mathbb{Z}_p\}.$$

Calculating the cardinality:

$$|F| = p^n.$$

□

2.6 Structure of Finite Fields

Lemma 2.41. *For every element β of a finite field F with q elements, we have:*

$$\beta^q = \beta.$$

Proof. If $\beta = 0$, then clearly $0^q = 0$.

Now assume $\beta \neq 0$. Let F^* denote the set of non-zero elements of F, written as:

$$F^* = \{\beta_1, \dots, \beta_{q-1}\}.$$

Consider the set formed by multiplying each element of F^* by β:

$$F^* = \beta F^* = \{\beta\beta_1, \dots, \beta\beta_{q-1}\}.$$

Since $\beta \neq 0$ and F is a field, the multiplication by β permutes the elements of F^*. Thus, the product of elements in the first set equals the product of elements in the second set:

$$\prod_{i=1}^{q-1} \beta_i = \prod_{i=1}^{q-1} (\beta\beta_i).$$

Factor out β from the product on the right-hand side (there are $q-1$ terms):

$$\prod_{i=1}^{q-1} \beta_i = \beta^{q-1} \prod_{i=1}^{q-1} \beta_i.$$

Since $\prod_{i=1}^{q-1} \beta_i$ is the product of non-zero elements, it is non-zero and can be cancelled from both sides:

$$1 = \beta^{q-1}.$$

Multiplying both sides by β, we obtain the result:

$$\beta = \beta^q.$$

□

Remark 2.42 (Connection between Roots and Divisibility)*.* An important connection between roots and divisibility exists for polynomials over a field F.

Statement: An element $\beta \in F$ is a root of $f \in F[x]$ if and only if $(x - \beta)$ divides $f(x)$.

Proof: By the polynomial division algorithm, we can write:

$$f(x) = q(x)(x - \beta) + c$$

with $q \in F[x]$ and $c \in F$ (the remainder is a constant because the divisor has degree 1).

Substituting $x = \beta$:

$$f(\beta) = q(\beta)(\beta - \beta) + c = c.$$

Therefore, the equivalence holds:

- If $(x - \beta) \mid f(x) \implies$ remainder $c = 0 \implies f(\beta) = 0$ (β is a root).
- If $f(\beta) = 0 \implies c = 0 \implies (x - \beta) \mid f(x)$.

Definition 2.43 (Multiplicity)**.** Let F be a subfield of E and $\beta \in E$ be a root of a nonzero polynomial $f(x) \in F[x]$.

The multiplicity of β in $f(x)$ is the largest integer m such that:

$$(x - \beta)^m \mid f(x)$$

A root is called simple if its multiplicity is 1.

Theorem 2.44. *Let $f(x) \in F[x]$ be of degree $n \geq 0$. Then $f(x)$ has at most n roots, counting multiplicity, in every extension field of F.*

Proof. Let $f(x) \in F[x]$ with $\deg(f) = n \geq 0$.

Let $\beta_1, \beta_2, \ldots$ be the distinct roots of f in some extension field E of F, with multiplicities $m_1, m_2, \ldots$ respectively.

Since each $(x - \beta_i)^{m_i}$ is a factor of $f(x)$ and the β_i are distinct, their product must divide $f(x)$:

$$\prod_{i \geq 1} (x - \beta_i)^{m_i} \mid f(x)$$

Now, consider the degree of the polynomial on the left side:

$$\deg \left(\prod_{i \geq 1} (x - \beta_i)^{m_i} \right) = \sum_{i \geq 1} m_i$$

Since the divisor's degree cannot exceed the dividend's degree:

$$\sum_{i \geq 1} m_i \leq \deg(f) = n$$

Therefore, f has at most n roots (counting multiplicity). □

Corollary 2.45. *Let F be a subfield of E with $|F| = q$. Then an element $\beta \in E$ lies in F if and only if:*

$$\beta^q = \beta$$

Proof. ($\implies$) All elements in F are roots of $x^q - x$ (from the previous Lemma). Thus, if $\beta \in F$, then $\beta^q - \beta = 0 \implies \beta^q = \beta$.

($\Longleftarrow$) Consider the polynomial $g(x) = x^q - x$. We know that $\deg(x^q - x) = q$, so $x^q - x$ has at most q roots in E.

Since $|F| = q$ and every element of F is a root of this polynomial, the set of roots of $x^q - x$ in E is exactly F.

Therefore, if $\beta^q = \beta$ (i.e., β is a root of $x^q - x$), then β must be in F.

$$\Longrightarrow \text{ All the roots of } x^q - x \text{ are in } F.$$

□

2.7 Uniqueness of Finite Fields

Definition 2.46 (Composite fields)**.** For any two finite fields E and F, the composite field $E \cdot F$ is the **smallest** field containing both E and F.

Theorem 2.47. *For any prime p and integer $n \geq 1$, there exists a **unique** finite field of p^n elements.*

Proof. **(Existence)** Let $f(x)$ be an irreducible polynomial of degree n over $\mathbb{Z}_p$. It follows that

$$\mathbb{Z}_p[x]/(f(x)) \quad \text{is a field.}$$

The size of $\mathbb{Z}_p[x]/(f(x))$ is p^n.

(Uniqueness) Let E, F be two finite fields of p^n elements. Consider the polynomial $x^{p^n} - x$ over the composite field $E \cdot F$.

Note that E is a subfield of $E \cdot F$. If $\beta \in E \cdot F$ and $\beta^{p^n} = \beta$, then $\beta \in E$.

We have the following conditions:

$$\begin{cases} x^{p^n} - x \text{ has at most } p^n \text{ roots.} \\ \text{All the elements in } E \text{ are roots of } x^{p^n} - x. \\ |E| = p^n \end{cases}$$

This implies:

$$E = \{\text{roots of } x^{p^n} - x \text{ in } E \cdot F\}$$

Similarly, we have:

$$F = \{\text{roots of } x^{p^n} - x \text{ in } E \cdot F\}$$

Therefore, $E = F$. □

Remark 2.48. The existence of such irreducible polynomial $f(x)$ should be justified.

Remark 2.49. 1. We can now denote the finite field with q elements by $\mathbb{F}_q$ (or $GF(q)$).

2. For an irreducible polynomial $f(x)$ with $\deg(f) = n$ over a field F, let α be a root of $f(x)$. The field $F[x]/(f(x))$ can be represented by

$$F(\alpha) := \{a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1} \mid a_i \in F\}$$

We will use $F(\alpha)$ to denote the field $F[x]/(f(x))$ if α is clear from the context.

3. $\mathbb{F}_p \cong \mathbb{Z}_p$, where p is a prime, while $\mathbb{F}_{p^n} \ncong \mathbb{Z}_{p^n} \quad (n > 1)$

2.8 Multiplicative Groups of a Finite Field

Definition 2.50 (Cyclic groups)**.** A group G is called cyclic if there is an element $a \in G$, called a generator, such that each element in G has the form a^n for some integer n.

Definition 2.51 (Primitive elements)**.** An element $\alpha \in \mathbb{F}_q$ is called a primitive element (or generator) of $\mathbb{F}_q$ if

$$\mathbb{F}_q = \{0, \alpha, \alpha^2, \ldots, \alpha^{q-1}\}.$$

Note that:

$$\mathbb{F}_q^* = \mathbb{F}_q \setminus \{0\} \quad \& \quad \alpha^{q-1} = 1 \quad \text{since} \quad \alpha^q = \alpha$$

Example 2.52. Let α be a root of the irreducible polynomial $f(x) = 1 + x + x^2 \in \mathbb{F}_2[x]$.

Consider $\mathbb{F}_4 = \mathbb{F}_2(\alpha) = \mathbb{F}_2[x]/(f(x))$.

$$\mathbb{F}_4 = \{0, 1, \alpha, 1 + \alpha\}$$

Since $f(\alpha) = 0 \implies 1 + \alpha + \alpha^2 = 0 \implies 1 + \alpha = -\alpha^2 = \alpha^2$ (in $\mathbb{F}_2$, $-1 = 1$).

Also:

$$\alpha^4 = \alpha \implies \alpha^3 = 1$$

Thus, we can rewrite the field elements as powers of α:

$$\mathbb{F}_4 = \{0, \alpha, \alpha^2, \alpha^3\}(\text{where} \quad \alpha^3 = 1)$$

Hence, α is primitive & $\mathbb{F}_4^*$ is cyclic.

Definition 2.53 (Order of elements in $\mathbb{F}_q$)**.** The order of a nonzero element $\alpha \in \mathbb{F}_q$, denoted by $\text{ord}(\alpha)$, is the smallest positive integer k s.t. $\alpha^k = 1$.

Example 2.54. $1 + x^2$ is irreducible over $\mathbb{F}_3 = \mathbb{Z}_3 = \{0, 1, 2\}$.

Let α be a root of $1 + x^2$.

Question: What is $\text{ord}(\alpha)$ in $\mathbb{F}_9$?

Solution: Consider $\mathbb{F}_9 = \mathbb{F}_3(\alpha) = \{0, 1, 2, \alpha, 1 + \alpha, 2 + \alpha, 2\alpha, 1 + 2\alpha, 2 + 2\alpha\}$.

We compute the powers of α:

$$\begin{aligned} 1+\alpha^2=0 \implies \alpha^2 &= -1 = 2 \neq 1 \\ \alpha^3 &= \alpha \cdot \alpha^2 = 2\alpha \neq 1 \\ \alpha^4 &= \alpha \cdot \alpha^3 = 2\alpha^2 = 4 = 1 \quad (\text{since } 4 \equiv 1 \pmod 3) \end{aligned}$$

$$\implies \mathrm{ord}(\alpha) = 4.$$

Is α primitive?
Answer: No. Note that $|\mathbb{F}_9| = 9$. A primitive element would have order $9-1=8$.

Note that $\{\alpha, 2, 2\alpha, 1\} = \{\alpha, \alpha^2, \alpha^3, \alpha^4\}$ is a cyclic group.

Lemma 2.55. *Let $\alpha \in \mathbb{F}_q^*$ be such that $\alpha^m = 1$ where m is a positive integer.*

(i) $\mathrm{ord}(\alpha) \mid m$. *In particular,* $\mathrm{ord}(\alpha) \mid q-1$.

(ii) For any $\alpha, \beta \in \mathbb{F}_q^*$, if $\gcd(\mathrm{ord}(\alpha), \mathrm{ord}(\beta)) = 1$, *then* $\mathrm{ord}(\alpha\beta) = \mathrm{ord}(\alpha)\mathrm{ord}(\beta)$.

Proof. (i) Write $m = a \cdot \mathrm{ord}(\alpha) + b$ where $a \geq 0$, $0 \leq b < \mathrm{ord}(\alpha)$. Then $1 = \alpha^m = \alpha^{a\cdot\mathrm{ord}(\alpha)+b} = \alpha^b$.

$$\implies b = 0. \quad \text{since } \mathrm{ord}(\alpha) \text{ is the smallest by definition}$$

$$\implies m = a \cdot \mathrm{ord}(\alpha) \implies \mathrm{ord}(\alpha) \mid m.$$

Note that $\alpha^{q-1} = 1$. We can take $m = q-1$. This shows $\mathrm{ord}(\alpha) \mid q-1$.

(ii) Denote $r = \mathrm{ord}(\alpha)\mathrm{ord}(\beta)$. Then

$$\alpha^r = \alpha^{\mathrm{ord}(\alpha)\mathrm{ord}(\beta)} = 1, \quad \beta^r = \alpha^{\mathrm{ord}(\alpha)\mathrm{ord}(\beta)} = 1.$$

Then we have

$$(\alpha\beta)^r = \alpha^r\beta^r = 1.$$

$$\overset{(i)}{\implies} \mathrm{ord}(\alpha\beta) \mid r, \quad \text{i.e., } \mathrm{ord}(\alpha\beta) \leq r.$$

On the other hand:

$$1 = (\alpha\beta)^{\mathrm{ord}(\alpha\beta)\cdot\mathrm{ord}(\alpha)} = \beta^{\mathrm{ord}(\alpha\beta)\mathrm{ord}(\alpha)}$$

$$\overset{(i)}{\implies} \mathrm{ord}(\beta) \mid \mathrm{ord}(\alpha\beta)\mathrm{ord}(\alpha).$$

Similarly,

$$\mathrm{ord}(\alpha) \mid \mathrm{ord}(\alpha\beta)\mathrm{ord}(\beta).$$

Since $\gcd(\mathrm{ord}(\alpha), \mathrm{ord}(\beta)) = 1$, we have

$$\mathrm{ord}(\alpha) \mid \mathrm{ord}(\alpha\beta) \quad \text{and} \quad \mathrm{ord}(\beta) \mid \mathrm{ord}(\alpha\beta).$$

$$\implies r \mid \operatorname{ord}(\alpha\beta).$$

$$\text{So } r \leq \operatorname{ord}(\alpha\beta).$$

Therefore, $\operatorname{ord}(\alpha\beta) = r$.

□

2.9 Characterization and Existence of Primitive Elements

Proposition 2.56. *(i) A nonzero element of $\mathbb{F}_q$ is a primitive element iff its order is $q-1$.*

(ii) Every finite field has at least one primitive element.

Proof. (i) The direction ($\Leftarrow$) follows immediately by the definition of the order.

For the direction ($\Rightarrow$), since $\alpha \in \mathbb{F}_q^*$, we know that $\alpha^{q-1} = 1$. By the definition of primitive elements, α generates the group, so its order must be exactly the size of the group, which is $q-1$.

(ii) Let m be the **least common multiple** of the orders of all elements in $\mathbb{F}_q^*$. We aim to show that $m = q-1$ and that there exists an element $\beta \in \mathbb{F}_q^*$ such that $\operatorname{ord}(\beta) = m$.

First, write the prime factorization of m as $m = p_1^{k_1} p_2^{k_2} \dots p_n^{k_n}$ for distinct primes $p_1, \dots, p_n$. By the definition of lcm, for all $i = 1, \dots, n$, there exists an element $\alpha_i \in \mathbb{F}_q^*$ such that $p_i^{k_i}$ divides $\operatorname{ord}(\alpha_i)$.

We define $\beta_i = \alpha_i^{\operatorname{ord}(\alpha_i)/p_i^{k_i}}$ for each i. We observe that:

$$(\beta_i)^{p_i^{k_i}} = \alpha_i^{\operatorname{ord}(\alpha_i)} = 1.$$

This implies $\operatorname{ord}(\beta_i) = p_i^{k_i}$.

Now, define $\beta = \prod_{i=1}^n \beta_i$. Since the orders $\operatorname{ord}(\beta_i) = p_i^{k_i}$ are powers of distinct primes, they are pairwise coprime, i.e., $\gcd(\operatorname{ord}(\beta_i), \operatorname{ord}(\beta_j)) = 1$ for $i \neq j$. Consequently, the order of the product is equal to the product of the orders:

$$\operatorname{ord}(\beta) = \prod_{i=1}^n \operatorname{ord}(\beta_i) = \prod_{i=1}^n p_i^{k_i} = m.$$

Since $\beta \in \mathbb{F}_q^*$, its order must divide the group order. Thus:

$$\operatorname{ord}(\beta) \mid (q-1) \implies m \mid (q-1) \implies m \leq q-1.$$

On the other hand, by the definition of m, the order of any element in $\mathbb{F}_q^*$ divides m. In other words, all elements in $\mathbb{F}_q^*$ are roots of the polynomial $x^m - 1$. Since the polynomial $x^m - 1$

has at most m roots in the field, and there are $q-1$ distinct elements in $\mathbb{F}_q^*$, we must have:

$$|\mathbb{F}_q^*| \leq m \implies q-1 \leq m.$$

Combining these inequalities, we conclude that $m = q-1$, then $ord(\beta) = q-1$

$\square$

Remark 2.57 (Finite Field Arithmetic). If α is a root of an irreducible polynomial of degree m over $\mathbb{F}_q$ and α is also **primitive** in $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^m}$, then every element in $\mathbb{F}_{q^m}$ can be represented both as a **polynomial** in α and a **power** of α.

That is:

$$\begin{aligned}\mathbb{F}_q(\alpha) = \mathbb{F}_{q^m} &= \{a_0 + a_1\alpha + \cdots + a_{m-1}\alpha^{m-1} \mid a_i \in \mathbb{F}_q\} \\ &= \{0, \alpha, \ldots, \alpha^{q^m-2}\}.\end{aligned}$$

- Addition: use the polynomial representation.
- Multiplication: use the power representation.

Example 2.58. Let $f(x) = x^3 + x + 1 \in \mathbb{F}_2[x]$. One can check that f is irreducible over $\mathbb{F}_2$.

Let α be a root of f. We define the extension field as:

$$\mathbb{F}_8 = \mathbb{F}_2(\alpha) = \mathbb{F}_2[x]/(f(x)).$$

Primitivity Check: Since $|\mathbb{F}_8^*| = 8 - 1 = 7$, the order of any nonzero element must divide 7.

$$\text{ord}(\alpha) \mid 7 \implies \text{ord}(\alpha) = 1 \text{ or } 7.$$

Since $\alpha \neq 1$, we conclude $\text{ord}(\alpha) = 7$, so α is primitive.

Element Representation: The elements of $\mathbb{F}_8$ can be expressed as powers of α or polynomials in α:

$$\begin{aligned}\mathbb{F}_8 &= \{0, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5, \alpha^6, \alpha^7\} \\ &= \{0, \alpha, \alpha^2, \alpha+1, \alpha^2+\alpha, \alpha^2+\alpha+1, \alpha^2+1, 1\}.\end{aligned}$$

Derivation Details: Using the relation $\alpha^3 + \alpha + 1 = 0 \implies \alpha^3 = \alpha + 1$, we compute the powers:

$$\begin{aligned}\alpha^3 &= \alpha + 1 \\ \alpha^4 &= \alpha(\alpha^3) = \alpha(\alpha+1) = \alpha^2 + \alpha \\ \alpha^5 &= \alpha(\alpha^4) = \alpha(\alpha^2+\alpha) = \alpha^3 + \alpha^2 = (\alpha+1) + \alpha^2 = \alpha^2 + \alpha + 1 \\ \alpha^6 &= \alpha(\alpha^5) = \alpha(\alpha^2+\alpha+1) = \alpha^3 + \alpha^2 + \alpha \\ &= (\alpha+1) + \alpha^2 + \alpha = \alpha^2 + 1 \\ \alpha^7 &= 1\end{aligned}$$

2.10 Vector Spaces

Definition 2.59. Let F be a field. Let $(V, +)$ be an additive abelian group (where the additive identity of V is denoted by 0) and let a function $F \times V \to V$ be defined satisfying that for any $\alpha, \beta \in F$ and $u, v \in V$:

(i) $1v = v$ (1 is the multiplicative identity of F)

(ii) $\alpha(\beta v) = (\alpha\beta)v$

(iii) $\alpha(u + v) = \alpha u + \alpha v$

(iv) $(\alpha + \beta)v = \alpha v + \beta v$

The operation $+$ is called **vector addition**, the function $F \times V \to V$ is called **scalar multiplication**, and V is called a **vector space over** F.

Example 2.60. Consider the set $V = \mathbb{F}_q^n$, which consists of all n-tuples with elements from the field $\mathbb{F}_q$:

$$V = \mathbb{F}_q^n := \{(v_1, \dots, v_n) \mid v_i \in \mathbb{F}_q\}.$$

We define the vector space operations as follows:

1. Addition (+):
For any two vectors $v = (v_1, \dots, v_n)$ and $w = (w_1, \dots, w_n)$ in $\mathbb{F}_q^n$, we define their sum component-wise:

$$v + w = (v_1 + w_1, \dots, v_n + w_n) \in \mathbb{F}_q^n.$$

2. Scalar Multiplication:
We define the mapping $\mathbb{F}_q \times \mathbb{F}_q^n \to \mathbb{F}_q^n$. For any scalar $\lambda \in \mathbb{F}_q$ and any vector $v \in \mathbb{F}_q^n$, we define:

$$\lambda v = (\lambda v_1, \dots, \lambda v_n).$$

Under these operations, $\mathbb{F}_q^n$ is a vector space over $\mathbb{F}_q$. (This structure is analogous to the real vector space $\mathbb{R}^n$ or the complex vector space $\mathbb{C}^n$).

Definition 2.61 (Subspaces)**.** A nonempty subset C of a vector space V is a **subspace** of V if it is itself a vector space with the same vector addition and scalar multiplication as V.

Example 2.62. 1. Let $0 \in \mathbb{F}_q^n$. Then $\{0\}$ is a subspace of $\mathbb{F}_q^n$.

2. Consider the set of vectors where the last $n - i$ coordinates are zero:

$$W = \{(v_1, \dots, v_i, 0, \dots, 0) \mid v_j \in \mathbb{F}_q\}.$$

Then W is a subspace of $\mathbb{F}_q^n$.

3. Let $1 \leq i < n$. Is $\mathbb{F}_q^i$ a subspace of $\mathbb{F}_q^n$?

No. The vectors in $\mathbb{F}_q^i$ have length i, while vectors in $\mathbb{F}_q^n$ have length n. Since the operations and elements must reside in the same space, $\mathbb{F}_q^i$ is not a subset of $\mathbb{F}_q^n$, and thus cannot be a subspace.

Proposition 2.63. *A nonempty subset C of a vector space V over $\mathbb{F}_q$ is a subspace iff the following is satisfied:*

$$\text{if } x, y \in C \text{ and } \lambda, \mu \in \mathbb{F}_q, \text{ then } \lambda x + \mu y \in C.$$

Proof. Omitted. □

Definition 2.64. A **linear combination** of $v_1, \ldots, v_n \in V$ over $\mathbb{F}_q$ is a vector of the form:

$$\lambda_1 v_1 + \cdots + \lambda_n v_n, \quad \text{where } \lambda_1, \ldots, \lambda_n \in \mathbb{F}_q.$$

Definition 2.65. A set of vectors $\{v_1, \ldots, v_n\} \subset V$ is **linearly independent** over $\mathbb{F}_q$ if

$$\lambda_1 v_1 + \cdots + \lambda_n v_n = 0 \implies \lambda_1 = \cdots = \lambda_n = 0.$$

The set is called **linearly dependent** if it is not linearly independent.

Definition 2.66 (Linear Span)**.** Let $S = \{v_1, \ldots, v_k\}$ be a nonempty subset of V. The **linear span** of S over $\mathbb{F}_q$ is defined as:

$$\langle S \rangle = \{\lambda_1 v_1 + \cdots + \lambda_k v_k \mid \lambda_i \in \mathbb{F}_q\}.$$

Remark 2.67.

1. If $S = \emptyset$, we define $\langle S \rangle = \{0\}$ (the zero vector of V).
2. The set $\langle S \rangle$ is a subspace of V (verify this). It is also called the **subspace generated** (or **spanned**) **by** S.
3. Given a subspace $C \subset V$, a subset $S \subset C$ is called a **generating** (or **spanning**) **set** for C if $C = \langle S \rangle$.

Example 2.68. Let $S \subset \mathbb{F}_2^4$ be given by:

$$S = \{0001, 1000, 1001\}.$$

The linear span is computed as:

$$\begin{aligned} \langle S \rangle &= \{\lambda_1(0001) + \lambda_2(1000) + \lambda_3(1001) \mid \lambda_1, \lambda_2, \lambda_3 \in \mathbb{F}_2\} \\ &= \{0001, 1000, 1001, 0000\}. \end{aligned}$$

Definition 2.69 (Basis)**.** (Plural form: **Bases**)

A nonempty subset $B = \{v_1, \ldots, v_k\}$ of a vector space V over $\mathbb{F}_q$ is called a **basis** for V if:

1. $V = \langle B \rangle$ (i.e., B spans V); and

2. B is linearly independent over $\mathbb{F}_q$.

Remark 2.70. (i) If $B = \{v_1, \ldots, v_k\}$ is a basis of V over $\mathbb{F}_q$, then for any $v \in V$ there exists **unique** $\lambda_1, \ldots, \lambda_k \in \mathbb{F}_q$ s.t.

$$v = \lambda_1 v_1 + \cdots + \lambda_k v_k.$$

(ii) A vector space V over $\mathbb{F}_q$ can have many bases.

E.g., if $\{v_1, v_2\}$ is a basis for $\mathbb{F}_2^2$, then $\{v_1 + v_2, v_2\}$ is also a basis for $\mathbb{F}_2^2$.

But all the bases have the same size.

Definition 2.71. The size of a basis (hence, any basis) of V over $\mathbb{F}_q$ is called the **dimension** of V over $\mathbb{F}_q$, denoted by $\dim(V)$ or $\dim_{\mathbb{F}_q}(V)$.

Theorem 2.72. *If* $\dim_{\mathbb{F}_q}(V) = k$*, then:*

(i) V has q^k elements.

(ii) V has $\frac{1}{k!}\prod_{i=0}^{k-1}(q^k - q^i)$ different bases.

Proof. **(i)** Let $\{v_1, \ldots, v_k\}$ be a basis for V over $\mathbb{F}_q$. Then:

$$V = \{\lambda_1 v_1 + \cdots + \lambda_k v_k \mid \lambda_i \in \mathbb{F}_q\}.$$

Since each coefficient λ_i can take q different values independently, the total number of elements is:

$$|V| = q^k.$$

(ii) Let us count the number of ways to construct a **sequence** of linearly independent vectors $v_1, \ldots, v_k$ in V.

- For v_1: By linear independence, $v_1 \neq 0$. There are $q^k - 1$ choices for v_1.
- For v_2: By linear independence, $v_2 \notin \langle v_1 \rangle$. Since $|\langle v_1 \rangle| = q$, there are $q^k - q$ choices for v_2.
- General step: For $2 \leq i \leq k$, we must have $v_i \notin \langle v_1, \ldots, v_{i-1} \rangle$. The subspace spanned by the previous $i-1$ vectors has q^{i-1} elements. So there are $q^k - q^{i-1}$ choices for v_i.

The total number of linearly independent **sequences** $v_1, \ldots, v_k$ is the product of these choices:

$$\prod_{i=1}^{k}(q^k - q^{i-1}) = \prod_{i=0}^{k-1}(q^k - q^i).$$

However, a basis is a **set**, not a sequence. The order of vectors does not matter. The number of permutations of the sequence $v_1, \ldots, v_k$ is $k!$.

Therefore, the number of different bases is:

$$\frac{1}{k!}\prod_{i=0}^{k-1}(q^k - q^i).$$

$\square$

Definition 2.73 (Scalar Product)**.** Let $v = (v_1, \ldots, v_n)$ and $w = (w_1, \ldots, w_n)$ be vectors in $\mathbb{F}_q^n$.

(i) The **scalar product** (or **dot product**) of v and w is defined as:

$$v \cdot w = v_1 w_1 + \cdots + v_n w_n.$$

(ii) The vectors v and w are said to be **orthogonal** if:

$$v \cdot w = 0.$$

(iii) Let S be a nonempty subset of $\mathbb{F}_q^n$. The **orthogonal complement** $S^\perp$ of S is defined to be:

$$S^\perp = \Big\{ v \in \mathbb{F}_q^n \mid v \cdot s = 0 \text{ for all } s \in S \Big\}.$$

If $S = \emptyset$, we define $S^\perp = \mathbb{F}_q^n$.

Remark 2.74. (i) $S^\perp$ is a subspace of $\mathbb{F}_q^n$.

(ii) $\langle S \rangle^\perp = S^\perp$.

(iii) Let $v \in \mathbb{R}^n$. Then $v \cdot v = 0$ implies $v = 0$.
But for $v \in \mathbb{F}_q^n$, $v \cdot v = 0$ **does not** imply $v = 0$.
Example: Consider $v = (1, 1) \in \mathbb{F}_2^2$. We have:

$$v \cdot v = 1 \cdot 1 + 1 \cdot 1 = 1 + 1 = 0, \quad \text{but } v \neq 0.$$

(Intuition: This implies that the scalar product doesn't induce a norm in $\mathbb{F}_q^n$).

Theorem 2.75. *Let S be a subset of $\mathbb{F}_q^n$. Then*

$$\dim(\langle S \rangle) + \dim(S^\perp) = n.$$

Proof. **Case 1:** If $S = \emptyset$, then by definition $\langle S \rangle = \{0\}$ and $S^\perp = \mathbb{F}_q^n$. Thus,

$$\dim(\{0\}) + \dim(\mathbb{F}_q^n) = 0 + n = n.$$

Case 2: Assume $S \neq \emptyset$. Let $k = \dim(\langle S \rangle)$. Let $\{v_1, \ldots, v_k\}$ be a basis for the subspace $\langle S \rangle$.

For any vector $x \in \mathbb{F}_q^n$, the condition $x \in S^\perp$ is equivalent to requiring that x is orthogonal to every basis vector of $\langle S \rangle$. That is:

$$x \in S^\perp \iff v_i \cdot x = 0 \quad \text{for all } i = 1, \ldots, k.$$

We can express this system of linear equations in matrix form. Let G be the $k \times n$ matrix whose rows are the basis vectors $v_1, \ldots, v_k$:

$$G = \begin{pmatrix} \text{---} & v_1 & \text{---} \\ \text{---} & v_2 & \text{---} \\ & \vdots & \\ \text{---} & v_k & \text{---} \end{pmatrix}.$$

Then $x \in S^\perp$ if and only if $Gx^T = 0$.

This implies that $S^\perp$ is exactly the **kernel** (or null space) of the linear transformation defined by G.

Since the rows of G are linearly independent (they form a basis), the **rank** of G is exactly k. By the **Rank-Nullity Theorem**:

$$\dim(\mathrm{Im}(G)) + \dim(\mathrm{Ker}(G)) = n.$$

Substituting the known values:

$$\mathrm{Rank}(G) + \dim(S^\perp) = n \implies k + \dim(S^\perp) = n.$$

Finally, since $\dim(\langle S \rangle) = k$, we have:

$$\dim(\langle S \rangle) + \dim(S^\perp) = n.$$

□

Example 2.76. Let $S = \{0100, 0101\} \subseteq \mathbb{F}_2^4$.

The linear span of S is:

$$\langle S \rangle = \{0000, 0100, 0101, 0001\}.$$

The set S is linearly independent over $\mathbb{F}_2$. Thus, we have:

$$\dim(\langle S \rangle) = 2.$$

Find $S^\perp$:

Let $x = (x_1, x_2, x_3, x_4) \in \mathbb{F}_2^4$. To find the orthogonal complement, x must be orthogonal to each vector in S. We can express this as a matrix equation:

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = 0$$

This yields the following system of linear equations:

$$\left.\begin{aligned} x_2 &= 0 \\ x_2 + x_4 &= 0 \end{aligned}\right\} \implies \begin{aligned} x_2 &= 0 \\ x_4 &= 0 \end{aligned}$$

Therefore, $S^\perp$ consists of all vectors in $\mathbb{F}_2^4$ where the second and fourth coordinates are zero:

$$\begin{aligned} S^\perp &= \{x \in \mathbb{F}_2^4 \mid x_2 = 0, x_4 = 0\} \\ &= \{0000, \mathbf{0010}, \mathbf{1000}, 1010\}. \end{aligned}$$

From this explicit set, we can easily identify that $\{\mathbf{0010}, \mathbf{1000}\}$ is a basis for $S^\perp$.

Consequently, we conclude:

$$\dim(S^\perp) = 2.$$

Chapter 3

Linear Codes

3.1 Linear Codes

Definition 3.1. A **linear code** C of length n over $\mathbb{F}_q$ is a subspace of $\mathbb{F}_q^n$.

The **dual code** of C, denoted by $C^\perp$, is the orthogonal complement of C in $\mathbb{F}_q^n$.

The **dimension** of C is the dimension of C as a vector space over $\mathbb{F}_q$.

Theorem 3.2. *Let $C \subset \mathbb{F}_q^n$ be a linear code over $\mathbb{F}_q$.*

(i) $|C| = q^{\dim(C)}$.

(ii) $C^\perp$ *is a linear code over* $\mathbb{F}_q$*, and*

$$\dim(C^\perp) + \dim(C) = n.$$

(iii) $(C^\perp)^\perp = C$.

Proof. (i) This follows immediately by the definition of linear codes and dimension.

(ii) Since C is a linear code (which is a subspace), it is equal to its own span, i.e., $\langle C \rangle = C$. Using the previously established theorem $\dim(C^\perp) + \dim(\langle C \rangle) = n$, we can substitute $\langle C \rangle$ with C to obtain:

$$\dim(C^\perp) + \dim(C) = n.$$

(iii) First, let us show that $C \subset (C^\perp)^\perp$. Let $c \in C$. For any $x \in C^\perp$, by the definition of the orthogonal complement, we have:

$$x \cdot c = 0.$$

This implies that $c \in (C^\perp)^\perp$. Therefore, we have established that $C \subset (C^\perp)^\perp$.

Next, we compare their dimensions. Applying the dimension formula from part (ii) to the subspace $C^\perp$, we get:

$$\dim((C^\perp)^\perp) + \dim(C^\perp) = n.$$

Substituting $\dim(C^{\perp}) = n - \dim(C)$ into the equation yields:

$$\dim((C^{\perp})^{\perp}) + (n - \dim(C)) = n \implies \dim((C^{\perp})^{\perp}) = \dim(C).$$

Since C is a subspace of $(C^{\perp})^{\perp}$ and they have the same dimension, they must contain the same number of elements:

$$|C| = |(C^{\perp})^{\perp}|.$$

Consequently, we conclude that:

$$C = (C^{\perp})^{\perp}.$$

□

Remark 3.3. (i) The size of a linear code must be a power of q.

(ii) An $(n, M = q^k, d)$ **linear** code over $\mathbb{F}_q$ is also called an $[n, k, d]$ **linear code** over $\mathbb{F}_q$.

3.2 Hamming Weight

Definition 3.4. For $x = (x_1, \ldots, x_n) \in \mathbb{F}_q^n$, the **Hamming weight** of x, denoted by $wt(x)$, is defined as:

$$wt(x) = d(x, 0), \quad \text{where } 0 \in \mathbb{F}_q^n.$$

Remark 3.5. 1. $wt(x) = \sum_{i=1}^{n} wt(x_i)$, where $wt(x_i) = 1$ iff $x_i \neq 0$, and 0 otherwise.

2. The set $Supp(x) := \{i \mid x_i \neq 0\}$ is called the **support** of x. It follows naturally that the weight is the size of the support:

$$wt(x) = |Supp(x)|.$$

Definition 3.6 (Symmetric Difference)**.** Let A and B be two sets. The **symmetric difference** of A and B, denoted by $A \Delta B$, is the set of elements that are in either A or B, but not in both.

Mathematically, it is defined as:

$$A \Delta B = (A \cup B) \setminus (A \cap B).$$

The number of elements in the symmetric difference satisfies the inclusion-exclusion principle:

$$|A \Delta B| = |A| + |B| - 2|A \cap B|.$$

Lemma 3.7. *If $x, y \in \mathbb{F}_q^n$, then $d(x, y) = wt(x - y)$.*

Proof. By applying the definition of Hamming weight and the coordinate-wise nature of the distance:

$$\begin{aligned} wt(x-y) &= d(x-y,0) \\ &= \sum_{i=1}^{n} d(x_i - y_i, 0) \\ &= \sum_{i=1}^{n} d(x_i, y_i) \\ &= d(x,y). \end{aligned}$$

□

Lemma 3.8. *If* $x, y \in \mathbb{F}_2^n$, *then*

$$wt(x+y) = wt(x) + wt(y) - 2wt(x * y)$$

where $x * y = (x_1 y_1, \ldots, x_n y_n)$.

Proof. First, we analyze the weight of the coordinate-wise product $x * y$. By definition, $x_i y_i \neq 0$ if and only if $x_i \neq 0$ and $y_i \neq 0$. In terms of support sets, this is equivalent to:

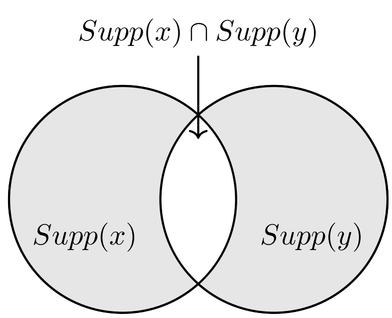
$$i \in Supp(x * y) \iff i \in Supp(x) \cap Supp(y).$$

Therefore, the weight of $x * y$ is the size of the intersection:

$$wt(x * y) = |Supp(x) \cap Supp(y)|.$$

Next, since we are working over $\mathbb{F}_2$, addition is performed modulo 2. A coordinate in the sum $x + y$ is non-zero if and only if exactly one of the corresponding coordinates in x or y is non-zero. This corresponds exactly to the **symmetric difference** (denoted by Δ) of their supports:

$$Supp(x+y) = Supp(x) \Delta Supp(y).$$



Using the inclusion-exclusion principle for the cardinality of the symmetric difference, we can expand the weight of $x+y$:

$$\begin{aligned} wt(x+y) &= |Supp(x) \Delta Supp(y)| \\ &= |Supp(x)| + |Supp(y)| - 2|Supp(x) \cap Supp(y)| \\ &= wt(x) + wt(y) - 2wt(x * y). \end{aligned}$$

This completes the proof. □

Lemma 3.9. *For any $x, y \in \mathbb{F}_q^n$, we have*

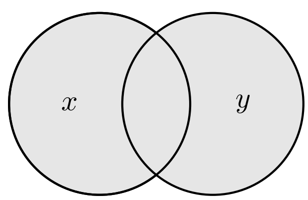
$$wt(x) + wt(y) \geq wt(x+y) \geq wt(x) - wt(y).$$

Proof. **Upper Bound:** The non-zero coordinates of $x+y$ must be non-zero in either x or y. Therefore, in terms of support sets:

$$Supp(x+y) \subseteq Supp(x) \cup Supp(y).$$

Taking the cardinality directly yields:

$$wt(x+y) \leq wt(x) + wt(y).$$



$Supp(x) \cup Supp(y)$

Lower Bound: By the triangle inequality and the translation invariance of Hamming distance:

$$d(x+y, 0) + d(y, 0) \geq d(x+y, y).$$

Since $d(x+y, y) = d(x, 0)$ and $d(y, 0) = d(0, y)$, we substitute the definition of weight:

$$wt(x+y) + wt(y) \geq wt(x).$$

Rearranging the terms yields:

$$wt(x+y) \geq wt(x) - wt(y).$$

□

3.3 Generator Matrix and Parity-Check Matrix

("Compact description" of linear codes)

Definition 3.10. (i) A **generator matrix** G of an $[n, k]$ linear code over $\mathbb{F}_q$ is a $k \times n$ matrix whose rows form a **basis** for C.

(ii) A **parity-check matrix** H for a linear code C is a **generator matrix for** $C^{\perp}$.

Remark 3.11. (i) A generator matrix of a given linear code is in general **not unique**.

(ii) By definition, the rows of a generator/parity-check matrix are **linearly independent**.

Example 3.12. (i) The $[n, 1]$ repetition code C over $\mathbb{F}_q$ is defined with a generator matrix:

$$G = (1, 1, \dots, 1)_{1 \times n}.$$

(ii) The $[n, n-1]$ parity code over $\mathbb{F}_q$ is defined with a parity-check matrix:

$$H = (1, 1, \dots, 1).$$

Lemma 3.13. *Let C be an $[n, k]$ linear code over $\mathbb{F}_q$ with generator matrix G, and let $v \in \mathbb{F}_q^n$. Then*

$$v \in C^{\perp} \iff Gv^T = 0.$$

*In particular, given an $(n-k) \times n$ matrix H, H is a **parity-check matrix** of C if and only if the rows of H are **linearly independent** and $GH^T = 0$.*

Proof. **Part 1: Prove** $v \in C^{\perp} \iff Gv^T = 0$

Write the generator matrix G as:

$$G = \begin{pmatrix} \text{---} & r_1 & \text{---} \\ & \vdots & \\ \text{---} & r_k & \text{---} \end{pmatrix},$$

where r_i is the i-th row of G.

($\Longrightarrow$) Assume $v \in C^{\perp}$. Since each row r_i is a codeword in C, by the definition of the orthogonal complement, we have $v \cdot r_i = 0$ for all $i = 1, \dots, k$. Therefore, $Gv^T = 0$.

($\Longleftarrow$) Assume $Gv^T = 0$, which means $v \cdot r_i = 0$ for all $i = 1, \dots, k$. For any codeword $c \in C$, we can write c as a linear combination of the basis rows: $c = \sum_{i=1}^{k} \lambda_i r_i$ for some $\lambda_i \in \mathbb{F}_q$. We then have:

$$v \cdot c = v \cdot \left(\sum_{i=1}^{k} \lambda_i r_i \right) = \sum_{i=1}^{k} \lambda_i (v \cdot r_i) = 0.$$

Since v is orthogonal to every codeword $c \in C$, we conclude $v \in C^{\perp}$.

Part 2: Prove H is a parity-check matrix $\iff$ rows of H are linearly independent and $GH^T = 0$

($\Longrightarrow$) Assume H is a parity-check matrix for C. By definition, the rows of H form a **basis** for $C^\perp$, and thus they are **linearly independent**. Moreover, each row of H is a codeword in $C^\perp$. By Part 1, since every row $h_j \in C^\perp$, we have $Gh_j^T = 0$. Combining these for all rows of H yields $GH^T = 0$.

($\Longleftarrow$) Assume $GH^T = 0$ and the $(n-k)$ rows of H are **linearly independent**. From $GH^T = 0$, applying Part 1 implies that each row of H is a codeword in $C^\perp$. Since there are $n-k$ linearly independent rows, they span a subspace of dimension $n-k$. We know from previous theorems that:

$$\dim(C^\perp) = n - \dim(C) = n - k.$$

Therefore, the $(n-k)$ rows of H form a **basis** for $C^\perp$. By definition, this means H is a parity-check matrix for C. □

Lemma 3.14. *Let C be an $[n,k]$ linear code over $\mathbb{F}_q$ with **parity-check matrix** H, and let $v \in \mathbb{F}_q^n$. Then*

$$v \in C \iff Hv^T = 0.$$

*In particular, given a $k \times n$ matrix G, G is a **generator** matrix of C iff the rows of G are **linearly independent** and $GH^T = 0$.*

Theorem 3.15 (Relation between $d(C)$ and H)**.** *Let C be a linear code and H be a parity-check matrix for C. Then*

*(i) $d(C) \geq d$ iff any $d-1$ columns of H are linearly **independent**.*

*(ii) $d(C) \leq d$ iff H has d columns that are linearly **dependent**.*

Proof. For any $v \in \mathbb{F}_q^n$, we have $v \in C \iff Hv^T = 0$.

Observation: There exist $d-1$ columns of H that are linearly **dependent** if and only if there exists $v \in \mathbb{F}_q^n \setminus \{0\}$ such that $wt(v) \leq d-1$ and $Hv^T = 0$, which is equivalent to saying there exists $v \in C \setminus \{0\}$ such that $wt(v) \leq d-1$.

(i) ($\Longrightarrow$) If $d(C) \geq d$, then $wt(C) \geq d$. This implies no $v \in C \setminus \{0\}$ is such that $wt(v) \leq d-1$. By our observation, it follows that any $d-1$ columns of H are linearly **independent**.

($\Longleftarrow$) Conversely, if any $d-1$ columns of H are linearly **independent**, then no $v \in C \setminus \{0\}$ is such that $wt(v) \leq d-1$. Thus, $d(C) \geq d$.

(ii) ($\Longrightarrow$) If $d(C) \leq d$, then there exists $v \in C \setminus \{0\}$ such that $1 \leq wt(v) \leq d$. So there exist d linearly **dependent** columns of H.

($\Longleftarrow$) Conversely, if there exist d **dependent** columns of H, then there exists $v \in \mathbb{F}_q^n \setminus \{0\}$ with $wt(v) \leq d$ such that $Hv^T = 0$. So $v \in C$ and $d(C) \leq wt(v) \leq d$.

□

Corollary 3.16. *Let C be a linear code and H be a parity-check matrix for C. Then $d(C) = d$ $\iff$ any $d-1$ columns of H are linearly* ***independent****, and H has d columns that are linearly* ***dependent****.*

Definition 3.17. (i) A generator matrix (for an $[n, k]$ linear code) of the form

$$(I_k, X)$$

where I_k is the $k \times k$ identity matrix, and X is some $k \times (n-k)$ matrix, is said to be **systematic** or in **standard form**.

(ii) A parity-check matrix (for an $[n, k]$ linear code) of the form

$$(Y, I_{n-k})$$

is said to be **systematic** or in **standard form**.

Theorem 3.18. *If $G = (I_k \mid X)$ is a generator matrix in* ***standard form*** *of an $[n, k]$ linear code C, then a parity-check matrix for C is*

$$H = \left(-X^T \mid I_{n-k}\right).$$

Proof. To verify that H is a parity-check matrix, we must show that $GH^T = 0$ and that the rows of H are linearly independent.

First, we calculate the matrix product:

$$\begin{aligned} GH^T &= (I_k \mid X)\left(-X^T \mid I_{n-k}\right)^T \\ &= (I_k \mid X)\begin{pmatrix} -X \\ I_{n-k} \end{pmatrix} \\ &= I_k(-X) + X(I_{n-k}) \\ &= -X + X \\ &= 0. \end{aligned}$$

Next, we observe that the rows of $H = \left(-X^T \mid I_{n-k}\right)$ are linearly **independent** because the rows of the identity block I_{n-k} are linearly **independent**.

Therefore, H is a valid parity-check matrix for C. □

Example 3.19. (i) The $[n, 1]$ repetition code C_R over $\mathbb{F}_q$ has the generator matrix:

$$G_R = \left(\underbrace{1}_{I_1} \mid \underbrace{1, \ldots, 1}_{X}\right)_{1 \times n}.$$

By the theorem, a parity-check matrix for C_R is $H_R = \left(-X^T \mid I_{n-1}\right)$, which takes the form:

$$H_R = \begin{pmatrix} -1 & 1 & & 0 \\ \vdots & & \ddots & \\ -1 & 0 & & 1 \end{pmatrix}_{(n-1)\times n}.$$

We can observe that any $n-1$ columns of H_R are linearly **independent**, but the n columns of H_R are **dependent**. This confirms that $d(C_R) = n$.

(ii) The $[n, n-1]$ parity code C_P over $\mathbb{F}_q$ has:

$$H_P = \left(\underbrace{1, \ldots, 1}_{-X^T} \mid \underbrace{1}_{I_1}\right)$$

as its parity-check matrix. From H_P, we can deduce that the distance is $d(C_P) = 2$.

Since $-X^T = (1, \ldots, 1)$, we have X as a column vector of -1s. Thus, the corresponding standard form generator matrix is $G_P = (I_{n-1} \mid X)$, which gives:

$$G_P = \begin{pmatrix} 1 & & & -1 \\ & \ddots & & \vdots \\ & & 1 & -1 \end{pmatrix}_{(n-1)\times n}.$$

Example 3.20. Consider the binary linear code C with generator matrix

$$G = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Since the rows are linearly independent, we have $\dim(C) = 2$.

To find the number of distinct bases for C, we use the dimension $k = 2$. The number of distinct bases is given by:

$$\frac{1}{2!}(2^2 - 2^0)(2^2 - 2^1) = \frac{1}{2}(3)(2) = 3.$$

We can explicitly list all distinct bases for C:

- $\{001, 100\}$
- $\{001, 101\}$
- $\{101, 100\}$

From these bases, we can write down all possible generator matrices of C by arranging the basis vectors as rows:

$$\begin{pmatrix} 0 & 0 & 1 \\ \mathbf{1} & \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 & 1 \\ \mathbf{1} & \mathbf{0} & \mathbf{1} \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 1 \\ \mathbf{1} & \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ \mathbf{1} & \mathbf{0} & \mathbf{1} \end{pmatrix}$$

Observe that **none** of these generator matrices is in the standard form.

If we permute the columns of G, we can obtain a standard form matrix:

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \xrightarrow{\text{permute cols.}} \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right).$$

However, permuting the columns of G does **not** produce a generator matrix for C in general.

Definition 3.21 (Equivalent codes)**.** Two (n, M) codes over $\mathbb{F}_q$ are **equivalent** (in the sense that n, M, d are the same) if one can be obtained from another by a combination of operations of the following type:

(i) **permutation** of the n symbols of the codewords;

(ii) **multiplication** of the symbols appearing in a fixed position by a nonzero element in $\mathbb{F}_q$.

Example 3.22. Let C be a $(3, 3)$ code over $\mathbb{F}_3$ given by

$$C = \{000, 011, 022\}.$$

We can obtain equivalent codes by applying the allowed operations to the codewords:

$$\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{array} \xrightarrow{\text{permute col 1 with col 2}} \begin{array}{ccc} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 2 & 0 & 2 \end{array} \xrightarrow{\text{multiply col 3 by 2}} \begin{array}{ccc} 0 & 0 & 0 \\ 1 & 0 & 2 \\ 2 & 0 & 1 \end{array}$$

Thus, C is equivalent to $C_1 = \{000, 101, 202\}$, which is equivalent to $C_2 = \{000, 102, 201\}$.

Theorem 3.23. *Any $[n, k]$ linear code C is* ***equivalent to*** *a linear code C' with a generator matrix in standard form.*

Example 3.24. The code $C = \{000, 011, 022\}$ is in fact a $[3, 1]$ linear code.

A generator matrix for C is:

$$G = \begin{pmatrix} 0 & 2 & 2 \end{pmatrix}.$$

By elementary row operations (multiplying the row by 2 in $\mathbb{F}_3$), we get:

$$\widetilde{G} = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix}.$$

By column permutation (swapping column 1 and column 2), we obtain:

$$G' = \begin{pmatrix} \mathbf{1} & \mathbf{0} & 1 \end{pmatrix}.$$

G' generates a code C' **equivalent to** C, and G' is successfully transformed into the standard form.

Proof of the theorem. Let G be a generator matrix for C.

Use elementary row operations to place G in the **Reduced Row Echelon Form** (RREF), i.e., there exists a $k \times k$ matrix A over $\mathbb{F}_q$ s.t.

$$\widetilde{G} = AG$$

and $\widetilde{G}$ is in RREF.

Then we can permute the columns of $\widetilde{G}$ so that the first k columns form the identity matrix I_k, i.e., there exists an $n \times n$ matrix B s.t.

$$G' := \widetilde{G}B = [I_k \mid X].$$

Clearly, G' generates a linear code equivalent to C. □

3.4 Encoding with a Linear Code

Definition 3.25. Let G be a generator matrix of an $[n, k]$ linear code C. Let r_i be the i-th row of G. For any message $u = (u_1, \ldots, u_k) \in \mathbb{F}_q^k$,

$$uG = u_1\mathbf{r}_1 + \cdots + u_k\mathbf{r}_k$$

is a codeword of C. Conversely, for any codeword $v \in C$, there exists a unique message $u \in \mathbb{F}_q^k$ s.t. $uG = v$.

Therefore,

$$C = \{uG \mid u \in \mathbb{F}_q^k\}.$$

The process of representing $u \in \mathbb{F}_q^k$ as a codeword $v = uG \in C$ is called **encoding**, where u is referred to as the **encoded message**.

Example 3.26. Let

$$G = \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right)$$

be a generator matrix over $\mathbb{F}_2$. It generates the $[3,2]$ parity code.

Let $u = (1,1) \in \mathbb{F}_2^2$.

Then

$$v = uG = (1,1)\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right) = (\underbrace{1,1}_{u},0).$$

In general,

$$v = u\,(I_k \mid X) = (u, uX).$$

3.5 Decoding of Linear Codes

Definition 3.27 (Cosets)**.** Let C be a linear code of length n over $\mathbb{F}_q$. We define the coset of C determined by $u \in \mathbb{F}_q^n$ to be

$$C + u := \{u + v \mid v \in C\}.$$

We may also denote $\underbrace{C+u}_{\text{right coset}}$ by $\underbrace{u+C}_{\text{left coset}}$.

Example 3.28. Let $C = \{000, 101, 010, 111\} \subseteq \mathbb{F}_2^3$. C is a linear code over $\mathbb{F}_2$.

For $u = 000$,

$$C + u = C.$$

For $u = 001$,

$$C + u = \{001, 100, 011, 110\}.$$

Theorem 3.29. *Let C be an $[n,k,d]$ linear code over $\mathbb{F}_q$. Then*

(i) every vector of $\mathbb{F}_q^n$ is in some coset of C.

(ii) for all $u \in \mathbb{F}_q^n$, $|C+u| = |C| = q^k$.

(iii) for all $u, v \in \mathbb{F}_q^n$, $u \in C + v$ implies $C + u = C + v$.

(iv) two cosets of C are either identical or have empty intersection.

(v) there are q^{n-k} cosets of C.

(vi) for all $u, v \in \mathbb{F}_q^n$, $u - v \in C$ iff u and v are in the same coset of C.

Remark 3.30. A vector $v \in C + u$ is called a **coset leader** if $wt(v) \leq wt(v')$ for all $v' \in C + u$.

Two methods for implementing minimum distance decoding of linear codes:

Standard array decoding

Example 3.31. Let $C = \{0000, 1011, 0101, 1110\} \subseteq \mathbb{F}_2^4$ be a $[4,2]$ linear code over $\mathbb{F}_2$.

We construct a 4×4 array of vectors of $\mathbb{F}_2^4$:

	coset leaders			
C	0000	1011	0101	1110
$C + 0001$	0001	1010	0100	1111
$C + 0010$	0010	1001	0111	1100
$C + 1000$	1000	0011	1101	0110

Suppose $c \in C$ was sent over a channel and the word $w = c + e$ is received, where $e \in \mathbb{F}_2^4$. Note that $w \in C + e$.

(i) Suppose $w = 1101$. Using the standard array, we can decode w to $c = 0101$.
By construction of the array, $w - c$ is the vector that has the **smallest** weight in $C + (w - c)$. Thus,

$$d(w, c) = wt(w - c) \leq wt(w - c') = d(w, c'),$$

for any $c' \in C$, i.e., any $w - c' \in C + (w - c)$.

(ii) Suppose $w = 1111$. Then we decode to $c = 1110$.

Remark 3.32. (i) For an $[n, k]$ linear code over $\mathbb{F}_q$, the size of its standard array is

$$q^{n-k} \times q^k.$$

(ii) If n is large, standard array decoding will be impractical.

(iii) The standard array is not unique.

Syndrome decoding

A more efficient way of implementing standard array decoding.

Definition 3.33 (Syndrome)**.** Let C be an $[n, k]$ linear code over $\mathbb{F}_q$ and H a parity-check matrix for C. For any $w \in \mathbb{F}_q^n$, the **syndrome** of w (with respect to H) is the word

$$s := wH^T \in \mathbb{F}_q^{n-k}.$$

Theorem 3.34. *For any $u, v \in \mathbb{F}_q^n$, we have*

(i) $(u + v)H^T = uH^T + vH^T$.

(ii) $uH^T = 0 \iff u \in C$.

(iii) $uH^T = vH^T \iff (u - v)H^T = 0 \iff u$ *and* v *are in the same coset of* C.

Example 3.35. Let $C = \{0000, 1011, 0101, 1110\} \subseteq \mathbb{F}_2^4$ be a $[4, 2]$ linear code. Let the parity-check matrix be:

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}.$$

We construct a table mapping coset leaders to their corresponding syndromes:

Coset leader	Syndrome
0000	00
0001	01
0010	10
1000	11

If the received word $w = 1101$, then

$$wH^T = (1,1,0,1) \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = 11.$$

We can decode w to:

$$w - (1000) = 0101.$$

If $w = 1111$, then $wH^T = 01$, and we can decode w to:

$$w - (0001) = 1110.$$

Chapter 4

Bounds of Codes

4.1 Bounds on the Parameters of Codes

Parameters of codes: n, M, d, q (where $M = q^k$ for linear codes, and $q = p^m$).

Example 4.1. Consider $C_1 = \{000, 111\} \subseteq \mathbb{F}_2^3$. For this code, $n = 3$, $M = 2$, $d = 3$, $q = 2$.
Consider $C_2 = \{000, 011, 101, 110\} \subseteq \mathbb{F}_2^3$. For this code, $n = 3$, $M = 4$, $d = 2$, $q = 2$.

Definition 4.2. For a given alphabet Q of size $q > 1$ and given positive integers n, d, let

$$A_q(n,d) := \max\{M \mid \text{there exists an } (n,M,d) \text{ code over } \mathbf{Q}\}.$$

In other words, $A_q(n,d)$ is **the largest possible size** M for which there exists an (n, M, d) code over Q.

Definition 4.3. For a given **prime power** q and given values of n, d, let

$$B_q(n,d) := \max\left\{q^k \mid \text{there exists an } [n,k,d] \text{ linear code over } \mathbb{F}_q\right\}.$$

Theorem 4.4. *Let $q \geq 2$ be a prime power. Then*

(i) $B_q(n,d) \leq A_q(n,d) \leq q^n$ *for all* $1 \leq d \leq n$.

(ii) $B_q(n,1) = A_q(n,1) = q^n$.

(iii) $B_q(n,n) = A_q(n,n) = q$.

Proof. (i) By definition, a linear code is a code, and any code is a subset of $\mathbb{F}_q^n$. Thus, $B_q(n,d) \leq A_q(n,d) \leq q^n$.

(ii) Let $C = \mathbb{F}_q^n$. Then C is an $[n, n, 1]$ linear code over $\mathbb{F}_q$. So

$$q^n \geq A_q(n,1) \geq B_q(n,1) \geq |C| = q^n \implies q^n = A_q(n,1) = B_q(n,1).$$

(iii) Let C be an (n, M, n) code over Q with $|Q| = q$. Since $d(C) = n$, all M codewords must differ in every coordinate. Restricting to the first coordinate, there can be at most q distinct symbols. Thus, $M \leq q \implies A_q(n,n) \leq q$.

Observe that the $[n,1,n]$ repetition code over $\mathbb{F}_q$ is linear and has size $M = q$. So $B_q(n,n) \geq q$. Because we always have $A_q(n,n) \geq B_q(n,n)$, combining these gives:

$$q \geq A_q(n,n) \geq B_q(n,n) \geq q.$$

Therefore, $B_q(n,n) = A_q(n,n) = q$.

□

4.2 Useful Quantities for Obtaining Bounds

Definition 4.5 (Hamming spheres/balls)**.** For any $u \in Q^n$ and integer $r \geq 0$, the sphere of radius r and center u is given by

$$S_Q(u,r) := \{v \in Q^n \mid d(u,v) \leq r\}.$$

Definition 4.6 (Volume of Hamming spheres)**.** The volume of $S_Q(u,r)$ is defined to be

$$V_q(n,r) = |S_Q(u,r)|.$$

Remark 4.7.

$$V_q(n,r) = \begin{cases} q^n & \text{if } r > n \\ \sum_{i=0}^{r}(q-1)^i\binom{n}{i} & \text{if } 0 \leq r \leq n \end{cases}$$

4.3 Lower Bounds on the Parameters

Theorem 4.8 (The sphere-covering bound/Gilbert-Varshamov(GV) bound for general code)**.** *For integers q,n,d s.t. $q > 1$, $1 \leq d \leq n$, we have*

$$A_q(n,d) \geq \frac{q^n}{V_q(n,d-1)}.$$

Proof. Let C be an optimal (n,M,d) code over an alphabet Q (where $|Q| = q$), meaning its size is maximal: $M = A_q(n,d)$.

We claim that the Hamming spheres of radius $d-1$ centered at all codewords $c \in C$ must completely cover the entire space Q^n.

To see why, suppose for the sake of contradiction that they do not cover Q^n. Then there must exist some word $w \in Q^n \setminus C$ that lies outside all these spheres. This implies that the distance from w to any existing codeword $c \in C$ is strictly greater than $d-1$, which means $d(w,c) \geq d$. If such a word w exists, we could add it to our code C to form a larger code with the exact same minimum distance d. However, this contradicts our initial assumption that C already has the maximum possible size $A_q(n,d)$.

Therefore, no such word w can exist, and the spheres $S_Q(c, d-1)$ for all $c \in C$ must indeed cover the entire space Q^n. We can express this set relationship as:

$$\bigcup_{c \in C} S_Q(c, d-1) \supset Q^n.$$

By taking the volume (the number of elements) of both sides, and noting that the volume of a union of sets is always less than or equal to the sum of their individual volumes, we obtain:

$$|C| \cdot V_q(n, d-1) \geq |Q^n| = q^n.$$

Since we know that $M = |C| = A_q(n, d)$, dividing both sides of the inequality by $V_q(n, d-1)$ directly yields our desired lower bound:

$$A_q(n, d) \geq \frac{q^n}{V_q(n, d-1)}.$$

□

Theorem 4.9 (The Gilbert-Varshamov(GV) bound for linear codes)**.** *Let q be a prime power and let n, k, d be positive integers such that $2 \leq d \leq n$, $1 \leq k \leq n$ and*

$$V_q(n-1, d-2) < q^{n-k}. \quad (*)$$

Then there exists an $[n, k]$ linear code over $\mathbb{F}_q$ with minimum distance at least d.

Proof. We construct a parity-check matrix H of size $(n-k) \times n$ column by column:

$$H = \begin{pmatrix} | & | & & | \\ h_1 & h_2 & \dots & h_n \\ | & | & & | \end{pmatrix}.$$

This matrix defines a code $C := \{v \in \mathbb{F}_q^n \mid Hv^T = 0\}$ with parameters $[n, \geq k]$.

For $1 \leq j \leq n$, let $h_j \in \mathbb{F}_q^{n-k}$ (where the total size is q^{n-k}) be chosen such that h_j is not in the linear span of **any** $d-2$ vectors of $\{h_1, \dots, h_{j-1}\}$. If such a choice is always possible, then any $d-1$ columns of H will be linearly independent, which implies that the code has minimum distance at least d.

Note that the number of vectors in the linear span of **at most** $d-2$ vectors of $\{h_1, \dots, h_{j-1}\}$ is bounded by:

$$\begin{aligned}\sum_{i=0}^{d-2} (q-1)^i \binom{j-1}{i} &= (q-1)^{d-2}\binom{j-1}{d-2} + (q-1)^{d-3}\binom{j-1}{d-3} + \dots + \binom{j-1}{0} \\ &= V_q(j-1, d-2).\end{aligned}$$

Since the volume of the Hamming sphere is monotonically increasing with the dimension, we have:

$$V_q(j-1, d-2) \leq V_q(n-1, d-2) < q^{n-k}.$$

Because the number of vectors in the forbidden linear span is strictly less than the total number of available vectors q^{n-k}, we can always find a valid h_j for **all** $j = 1, \ldots, n$.

Therefore, H successfully defines a linear code with minimum distance at least d. □

Corollary 4.10. *For $2 \leq d \leq n$, we have*

$$B_q(n, d) \geq q^{n-\lceil \log_q(V_q(n-1,d-2)+1) \rceil} \geq \frac{q^{n-1}}{V_q(n-1, d-2)}.$$

Proof. Take $k = n - \lceil \log_q(V_q(n-1, d-2) + 1) \rceil$.

This choice of k implies that $n - k = \lceil \log_q(V_q(n-1, d-2) + 1) \rceil$, which guarantees that

$$q^{n-k} \geq V_q(n-1, d-2) + 1 > V_q(n-1, d-2).$$

Then, by the GV bound theorem, there exists an $[n, k]$ linear code $\widetilde{C}$ with minimum distance at least d. Let $d(\widetilde{C}) \geq d$ be its exact minimum distance.

Let $c \in \widetilde{C}$ be a minimum weight codeword, so $wt(c) = d(\widetilde{C}) \geq d$. Let $I \subset \text{Supp}(c) = \{i \mid c_i \neq 0\}$ be a subset of the support of c with size

$$|I| = d(\widetilde{C}) - d.$$

For all codewords in $\widetilde{C}$, change the symbols on the coordinates in I to zeros. Since the minimum weight of any non-zero codeword in $\widetilde{C}$ is $d(\widetilde{C})$, and we only zeroed out $|I| < d(\widetilde{C})$ coordinates, no non-zero codeword becomes the zero vector. Thus, the linear independence of the basis is preserved.

The resulting code $\widehat{C}$ is still an $[n, k]$ linear code. Furthermore, the weight of the specific codeword c is reduced by exactly $|I|$, yielding a new weight of $d(\widetilde{C}) - (d(\widetilde{C}) - d) = d$. Since no codeword's weight can drop by more than $|I|$, the new minimum distance is exactly d.

Therefore, $\widehat{C}$ is an $[n, k, d]$ linear code. It follows that the maximum size of such a code is bounded by:

$$B_q(n, d) \geq |\widehat{C}| = q^k = q^{n-\lceil \log_q(V_q(n-1,d-2)+1) \rceil}.$$

Using the inequality $\lceil \log_q(x + 1) \rceil \leq \log_q(x) + 1$ for integer x, we can further loosen the bound,

$$q^{n-\lceil \log_q(V_q(n-1,d-2)+1) \rceil} \geq q^{n-1-\log_q(V_q(n-1,d-2))} = \frac{q^{n-1}}{V_q(n-1, d-2)}.$$

□

4.4 Upper Bounds (Necessary Conditions for Existence of Codes)

Theorem 4.11 (The Hamming bound/sphere-packing bound)**.** *For any integer $q > 1$ and n, d such that $1 \leq d \leq n$, we have*

$$A_q(n,d) \leq \frac{q^n}{V_q\left(n, \lfloor \frac{d-1}{2} \rfloor\right)}.$$

Proof. Let C be an (n, M, d) code over $Q = \{0, 1, \ldots, q-1\}$.

The **Hamming spheres** in Q^n of radius $r = \lfloor \frac{d-1}{2} \rfloor$ that are centered at codewords of C must be **disjoint**.

Thus, the union of these spheres is a subset of the entire space Q^n:

$$\bigcup_{c \in C} S_Q(c, r) \subset Q^n.$$

Since the spheres are disjoint, the volume of their union is the sum of their individual volumes. Taking the cardinality gives:

$$M \cdot V_q(n,r) = |C| V_q(n,r) = \left| \bigcup_{c \in C} S_Q(c,r) \right| \leq q^n.$$

Therefore, we obtain the upper bound for the size of the code:

$$M \leq \frac{q^n}{V_q(n,r)}$$

for any q-ary (n, M, d) code. Since $A_q(n,d)$ is the maximum possible value for M, the theorem follows. $\square$

Definition 4.12 (Perfect codes)**.** A code is called perfect if it attains the Hamming bound.

Definition 4.13 (Hamming codes)**.** Let $r \geq 2$ and q be a prime power. A linear code over $\mathbb{F}_q$ whose parity-check matrix H has the property that the columns of H are made up of precisely one nonzero vector from each vector space of dimension one of $\mathbb{F}_q^r$ is called a q-ary Hamming code, denoted by $\mathrm{Ham}(r, q)$.

Example 4.14. $\mathrm{Ham}(3,2)$ is a $[7,4]$ linear code over $\mathbb{F}_2$.

The parity-check matrix is:

$$H = \left(\begin{array}{ccc|cccc} 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right)$$

The columns correspond exactly to the elements of $\mathbb{F}_2^3 \setminus \{0\}$, and the number of rows is 3.

For this code, the minimum distance is $d = 3$.

Note that for each $v_1, v_2 \in \mathbb{F}_2^3$,

$$\langle v_1 \rangle = \langle v_2 \rangle \iff v_1 = v_2.$$

Remark 4.15. (i) The rows of the defining parity-check matrix H of $\mathrm{Ham}(r,q)$ are linearly independent since it contains all the r vectors of weight 1 in $\mathbb{F}_q^r$.

(ii) For any $u, v \in \mathbb{F}_q^r \setminus \{0\}$,
$$\langle u \rangle = \langle v \rangle \iff u = \lambda v, \lambda \in \mathbb{F}_q^*.$$

(iii) There are $\frac{q^r-1}{q-1}$ distinct one-dimensional subspaces of $\mathbb{F}_q^r$.

(iv) The dimension of H is
$$r \times \frac{q^r-1}{q-1}.$$

(v) When $q = 2$, H is an $r \times (2^r - 1)$ matrix. The columns of H in this case are all the nonzero vectors in $\mathbb{F}_2^r$.

(vi) The order of the columns of H has not been fixed in the definition. So $\mathrm{Ham}(r,q)$ is only defined up to code equivalence.

Proposition 4.16 (Properties of q-ary Hamming codes). *(i) Ham*(r,q) *is a* $\left[\frac{q^r-1}{q-1}, \frac{q^r-1}{q-1} - r, 3\right]$ *linear code.*

(ii) Ham(r,q) *is a perfect code.*

Proof. (i) Let $n = \frac{q^r-1}{q-1}$. The dimension of $\mathrm{Ham}(r,q)$ is $k = n - r = \frac{q^r-1}{q-1} - r$.

Any two columns of the parity-check matrix H are linearly independent by the definition of the Hamming code. Thus, $d(\mathrm{Ham}(r,q)) \geq 3$.

At the same time, H contains columns of the form $(\alpha, 0, \dots, 0)^T$ (with $r-1$ zeros at the bottom) and $(\beta_1, \beta_2, 0, \dots, 0)^T$ (with $r-2$ zeros at the bottom) for some $\alpha, \beta_1, \beta_2 \in \mathbb{F}_q^*$. These columns are linearly dependent. So $d(\mathrm{Ham}(r,q)) \leq 3$.

Combining these, we conclude that the minimum distance is exactly $d = 3$.

(ii) By the Hamming bound (sphere-packing bound) for $d = 3$, we have:

$$|\mathrm{Ham}(r,q)| \leq A_q(n,3) \leq \frac{q^n}{V_q(n,1)}.$$

We calculate the volume of the Hamming sphere of radius 1:

$$V_q(n,1) = \sum_{i=0}^{1} \binom{n}{i} (q-1)^i = \binom{n}{0}(q-1)^0 + \binom{n}{1}(q-1)^1 = 1 + n(q-1).$$

Since $n = \frac{q^r-1}{q-1}$, we have $n(q-1) = q^r - 1$. Substituting this into the volume equation gives:

$$V_q(n,1) = 1 + (q^r - 1) = q^r.$$

Therefore, the upper bound becomes:

$$A_q(n,3) \leq \frac{q^n}{q^r} = q^{n-r}.$$

Since the actual size of the Hamming code is exactly $|\mathrm{Ham}(r,q)| = q^k = q^{n-r}$, we achieve the equality:

$$A_q(n,3) = |\mathrm{Ham}(r,q)|.$$

Thus, $\mathrm{Ham}(r,q)$ attains the Hamming bound and is a perfect code.

□

4.5 Decoding with a Hamming Code

Let H be a parity-check matrix of $\mathrm{Ham}(r,q)$ and $w = c+e$ be the received word, where $c \in \mathrm{Ham}(r,q)$ is the transmitted codeword and e is the error vector.

Suppose $wt(e) \leq 1$, and let the error be:

$$e = e_{j,b} := (\underbrace{0,\ldots,0}_{j-1}, \underbrace{b}_{\text{magnitude}}, \underbrace{0,\ldots,0}_{n-j})$$

where j is the error location and b is the error magnitude.

We compute the syndrome wH^T:

$$wH^T = (c + e_{j,b})H^T = \underbrace{cH^T}_{0} + e_{j,b}H^T = e_{j,b}H^T.$$

Case 1: If $wH^T = 0$, then $e_{j,b}H^T = 0$, which equivalently means $He_{j,b}^T = 0$. Expressing H by its columns $(h_1, h_2, \ldots, h_n)$, we have:

$$(h_1, h_2, \ldots, h_n)\begin{pmatrix} 0 \\ \vdots \\ b \\ \vdots \\ 0 \end{pmatrix} = b \cdot h_j = 0.$$

Since h_j is a column of a Hamming code parity-check matrix, $h_j \neq 0$. So $b = 0$, and there is no error. We decode w to w.

Case 2: If $wH^T \neq 0$, then $e_{j,b}H^T \neq 0$. This evaluates to:

$$e_{j,b}H^T = b \cdot h_j^T \neq 0.$$

So we find the unique $e_{j,b}$ such that $wH^T = e_{j,b}H^T$ by inspecting the columns of H. We then decode w to $w - e_{j,b}$.

Example 4.17. Consider the Hamming code $\mathrm{Ham}(3,2)$ over $\mathbb{F}_2$ with the parity-check matrix:

$$H = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

Suppose the received word is $w = (1,0,0,1,0,0,1)$.

First, we compute the syndrome wH^T:

$$wH^T = (0,1,0).$$

We know that $wH^T = e_{j,b}H^T = b \cdot h_j^T = (0,1,0)$. Since the code is over $\mathbb{F}_2$, the non-zero error magnitude must be $b = 1$. This directly yields the equation:

$$h_j^T = (0,1,0).$$

By inspecting the columns of the parity-check matrix H, we find that $(0,1,0)^T$ corresponds precisely to the second column, i.e., $h_j = h_2$. This definitively locks the error location at $j = 2$.

The error vector is therefore:

$$e_{2,1} = (0,1,0,0,0,0,0).$$

Finally, we decode w by subtracting the error vector (which is equivalent to addition over $\mathbb{F}_2$):

$$w - e_{2,1} = (1,0,0,1,0,0,1) - (0,1,0,0,0,0,0) = (1,1,0,1,0,0,1).$$

4.6 Singleton Bound and MDS Codes

Theorem 4.18 (The Singleton bound)**.** *For any integer $q > 1$, $1 \le d \le n$, we have*

$$A_q(n,d) \le q^{n-d+1}.$$

In particular, when q is a prime power, any $[n,k,d]$ linear code over $\mathbb{F}_q$ satisfies

$$k \le n - d + 1.$$

Proof. Let C be an (n,M,d) code over Q, with $|Q| = q$.

Since $d(C) = d$, we have

$$d(c_1,c_2) \ge d \quad \text{for } c_1,c_2 \in C, c_1 \ne c_2.$$

Deleting the last $d-1$ coordinates of all codewords in C results in words of length $n-d+1$

that are all distinct.

Thus, the number of codewords M is bounded by the total number of possible words of length $n-d+1$:

$$M \leq q^{n-d+1}.$$

Therefore,

$$A_q(n,d) \leq q^{n-d+1}.$$

If C is an $[n,k,d]$ linear code over $\mathbb{F}_q$, then

$$|C| = M = q^k.$$

So,

$$q^k \leq q^{n-d+1}, \text{ i.e., } k \leq n-d+1.$$

□

Alternative proofs for linear codes. (i) Let H be a parity-check matrix. We know H has full rank $n-k$. Because the columns of H are vectors in $\mathbb{F}_q^{n-k}$, any set of more than $n-k$ columns must be linearly dependent. Therefore, there exist $n-k+1$ columns of H that are linearly dependent. Since the minimum distance d equals the minimum number of linearly dependent columns of H, it immediately follows that:

$$d \leq n-k+1.$$

(ii) Without loss of generality, assume the generator matrix G is in the standard form:

$$G = (I_k \mid X)$$

where the identity matrix I_k has size $k \times k$, and X has $n-k$ columns. By inspecting I_k, there are exactly $k-1$ zeros in the first k coordinates of each row of G. Thus, there are at least $k-1$ zeros in each row of G overall. Therefore, C has a codeword (each row of G is a valid codeword) of weight at most:

$$n-(k-1) = n-k+1.$$

Because the minimum distance d of a linear code is bounded by the weight of any non-zero codeword, we have:

$$d \leq n-k+1.$$

□

Definition 4.19 (Maximum distance separable (MDS) codes)**.** An (n,M,d) code over Q of size q

is called MDS if it attains the Singleton bound, namely, it satisfies

$$d = n - \log_q M + 1.$$

(For linear MDS codes, $d = n - k + 1$.)

Theorem 4.20 (Properties of MDS codes)**.** *Let C be an $[n, k < n, d]$ linear code over $\mathbb{F}_q$. Let G, H be a generator matrix and a parity-check matrix for C, respectively.*

The following statements are equivalent:

(i) C is MDS.

(ii) Every set of $n - k$ columns of H is linearly independent.

(iii) Every set of k columns of G is linearly independent.

(iv) $C^{\perp}$ is MDS.

Proof. **Proof of (i) $\iff$ (ii):**

(i) $\implies$ (ii) If C is MDS, then its minimum distance is $d = n - k + 1$. By the properties of the parity-check matrix H, a minimum distance of d implies that every $d - 1$ columns of H are linearly independent. Since $d - 1 = (n - k + 1) - 1 = n - k$, every set of $n - k$ columns of H is linearly independent.

(ii) $\implies$ (i) If every set of $n - k$ columns of H is linearly independent, then any linear dependence requires at least $n - k + 1$ columns. This means the minimum distance is $d(C) \geq n - k + 1$. At the same time, the Singleton bound states that $d(C) \leq n - k + 1$. Combining these gives $d(C) = n - k + 1$, hence C is MDS.

Proof of (iii) $\iff$ (iv):

The generator matrix G of C is exactly a parity-check matrix for the dual code $C^{\perp}$. By applying the equivalence (i) $\iff$ (ii) to $C^{\perp}$, the statement follows immediately.

Proof of (i) $\iff$ (iv):

(i) $\implies$ (iv) H is a generator matrix for $C^{\perp}$. So $C^{\perp}$ is an $[n, n - k]$ linear code.

By the Singleton bound, the minimum distance of $C^{\perp}$ is bounded by:

$$d(C^{\perp}) \leq n - (n - k) + 1 = k + 1.$$

Suppose for contradiction there exists a non-zero codeword $c^* \in C^{\perp} \setminus \{0\}$ such that $wt(c^*) \leq k$. Without loss of generality (WLOG), assume the last $n - k$ coordinates of c^* are zeros.

We can partition H into $H = (H_1, H_2)$, where H_1 is an $(n - k) \times k$ matrix and H_2 is an $(n - k) \times (n - k)$ matrix. Since we established (ii) holds (as C is MDS), every set of $n - k$ columns of H is linearly independent. Thus, H_2 has full rank and is invertible.

Since $c^* \in C^{\perp}$ and H generates $C^{\perp}$, there exists a vector x such that $xH = c^*$. This yields:

$$x(H_1, H_2) = (c_1^*, c_2^*) = (c_1^*, 0).$$

This implies $xH_2 = 0$. Since H_2 is invertible, we must have $x = 0$.

However, if $x = 0$, then $xH = 0 = c^*$, which contradicts our assumption that $c^* \neq 0$.

Hence, $wt(c^*) \geq k+1$ for all non-zero c^*. This means $d(C^\perp) = k+1$, which proves $C^\perp$ achieves the Singleton bound and is MDS.

(iv) $\implies$ **(i)** If $C^\perp$ is MDS, we can apply the proven implication (i) $\implies$ (iv) to $C^\perp$ itself. This implies that $(C^\perp)^\perp$ is MDS. Since $(C^\perp)^\perp = C$, it follows that C is MDS. □

Example 4.21. (i) $\mathbb{F}_q^n$ is an $[n, n, 1]$ linear code.

Comparing k versus $n - d + 1$, we have:

$$n = n - 1 + 1.$$

Thus, $\mathbb{F}_q^n$ is MDS.

(ii) The repetition code is an $[n, 1, n]$ code.

Its generator matrix is:

$$G = (\underbrace{1, \ldots, 1}_{n}).$$

Checking the Singleton bound condition ($k = n - d + 1$), we have:

$$1 = n - n + 1.$$

Thus, the repetition code is an MDS code.

(iii) The parity code is an $[n, n-1, 2]$ code.

Its parity-check matrix is:

$$H = (\underbrace{1, \ldots, 1}_{n}).$$

Checking the Singleton bound condition ($k = n - d + 1$), we have:

$$n - 1 = n - 2 + 1.$$

Thus, the parity code is an MDS code.

4.7 Reed-Solomon(RS) Codes

Definition 4.22. Let q, n, k be integers such that $0 \leq k \leq n \leq q$, where q is a prime power. Let $\alpha = (\alpha_1, \ldots, \alpha_n)$ be a vector of n **distinct** elements from $\mathbb{F}_q$. Let $u = (u_0, \ldots, u_{k-1}) \in \mathbb{F}_q^k$.

Associate a polynomial $u(x)$ with u as follows:

$$u(x) = \sum_{i=0}^{k-1} u_i x^i \in \mathbb{F}_q[x].$$

The RS code is defined to be

$$RS_q(\alpha, k) = \left\{ \underbrace{(u(\alpha_1), \dots, u(\alpha_n))}_{\in \mathbb{F}_q^n} \;\middle|\; u(x) \in \mathbb{F}_q[x], \deg u \leq k-1 \right\}.$$

Remark 4.23. (i) $\alpha_1, \dots, \alpha_n$ are called evaluation points / code locators.

(ii) The encoding of the message word $u \in \mathbb{F}_q^k$ under $RS_q(\alpha, k)$ is the evaluations of the polynomial $u(x)$ at $\alpha_1, \dots, \alpha_n$.

Namely, the encoding mapping can be defined by

$$\begin{aligned} \mathbb{F}_q^k &\longrightarrow \mathbb{F}_q^n \\ (u_0, \dots, u_{k-1}) &\longmapsto (u(\alpha_1), \dots, u(\alpha_n)). \end{aligned}$$

(iii) We may take $\alpha_i = \gamma^i$ where γ is primitive in $\mathbb{F}_q$.

(So $\alpha_1, \dots, \alpha_n$ are distinct since $q \geq n$).

Example 4.24. Consider $\mathbb{F}_3 = \{0, 1, 2\}$.

Let $n = 3$, $k = 2$, and $\alpha = (0, 1, 2)$.

$\mathbb{F}_3^2$ $u = (u_0, u_1)$	$\mathbb{F}_3[x]$	$RS_3((0,1,2),2)$
$(0,0)$	0	$(0,0,0)$
$(0,1)$	x	$(0,1,2)$
$(0,2)$	$2x$	$(0,2,1)$
$(1,0)$	1	$(1,1,1)$
$(1,1)$	$1+x$	$(1,2,0)$
$(1,2)$	$1+2x$	$(1,0,2)$
$(2,0)$	2	$(2,2,2)$
$(2,1)$	$2+x$	$(2,0,1)$
$(2,2)$	$2+2x$	$(2,1,0)$

Theorem 4.25. *$RS_q(\alpha, k)$ is an $[n, k, d = n-k+1]$ linear code over $\mathbb{F}_q$.*

Proof. First, we show that $RS_q(\alpha, k)$ is linear. Let $c_1 = (u_1(\alpha_1), \dots, u_1(\alpha_n))$ and $c_2 = (u_2(\alpha_1), \dots, u_2(\alpha_n))$, where $u_1, u_2 \in \mathbb{F}_q^k$. For any scalars $\lambda, \mu \in \mathbb{F}_q$, we have:

$$\begin{aligned} \lambda c_1 + \mu c_2 &= (\lambda u_1(\alpha_1) + \mu u_2(\alpha_1), \dots, \lambda u_1(\alpha_n) + \mu u_2(\alpha_n)) \\ &= ((\lambda u_1 + \mu u_2)(\alpha_1), \dots, (\lambda u_1 + \mu u_2)(\alpha_n)). \end{aligned}$$

Note that the linear combination of the polynomials is:

$$\lambda u_1 + \mu u_2 = \sum_{i=0}^{k-1} (\lambda u_{1,i} + \mu u_{2,i}) x^i.$$

Thus, $(\lambda u_1 + \mu u_2)(x)$ is a polynomial of degree at most $k-1$. Therefore, $\lambda c_1 + \mu c_2 \in RS_q(\alpha, k)$. So $RS_q(\alpha, k)$ is linear.

Next, we determine the minimum distance d. Recall that a nonzero polynomial $f(x) \in \mathbb{F}_q[x]$ of degree t has at most t distinct roots in $\mathbb{F}_q$. Let $u \in \mathbb{F}_q^k \setminus \{0\}$. The codeword corresponding to u is $(u(\alpha_1), \ldots, u(\alpha_n)) \neq 0$. The weight of this codeword is:

$$wt((u(\alpha_1), \ldots, u(\alpha_n))) = n - |\{\alpha_i \mid u(\alpha_i) = 0\}|.$$

Since $\deg u \leq k-1$, the polynomial $u(x)$ has at most $k-1$ distinct roots. This means $|\{\alpha_i \mid u(\alpha_i) = 0\}| \leq k-1$. This implies:

$$wt((u(\alpha_1), \ldots, u(\alpha_n))) \geq n - (k-1) = n - k + 1.$$

Because the minimum distance d of a linear code is the minimum weight of its non-zero codewords, we have $d \geq n - k + 1$. At the same time, by the Singleton bound, we know:

$$d \leq n - k + 1.$$

Thus, we conclude that $d = n - k + 1$.

Since $d \geq 1$, the mapping

$$\begin{aligned} RS_q(\alpha, k) : \mathbb{F}_q^k &\longrightarrow \mathbb{F}_q^n \\ u &\longmapsto (u(\alpha_1), \ldots, u(\alpha_n)) \end{aligned}$$

is injective. Therefore, $\dim(RS_q(\alpha, k)) = k$. □

Example 4.26 (Examples cont.)**.** Consider $\mathbb{F}_3 = \{0, 1, 2\}$.

Let $n = 3$, $k = 2$, and $\alpha = (0, 1, 2)$.

Since $RS_3((0,1,2),2)$ is a linear code by the above theorem, we can find a generator matrix for it. Observe $RS_3((0,1,2),2)$ is a $[3,2,2]$ linear code.

A generator matrix for $RS_3((0,1,2),2)$ is

$$G = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Remark 4.27. A generator matrix for $RS_q(\alpha, k)$ is

$$G_{RS} = \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{k-1} & \alpha_2^{k-1} & \dots & \alpha_n^{k-1} \end{pmatrix}.$$

This is a Vandermonde matrix of dimension $k \times n$.

Proof. To find the generator matrix G:

$$\begin{aligned} uG &= (u(\alpha_1), \dots, u(\alpha_n)) \\ (u_0, \dots, u_{k-1})G &= \left(\sum_{i=0}^{k-1} u_i \alpha_1^i, \dots, \sum_{i=0}^{k-1} u_i \alpha_n^i \right) \\ &= \left((u_0, \dots, u_{k-1}) \begin{pmatrix} \alpha_1^0 \\ \vdots \\ \alpha_1^{k-1} \end{pmatrix}, \dots, (u_0, \dots, u_{k-1}) \begin{pmatrix} \alpha_n^0 \\ \vdots \\ \alpha_n^{k-1} \end{pmatrix} \right) \\ &= (u_0, \dots, u_{k-1}) \underbrace{\begin{pmatrix} \alpha_1^0 & \alpha_2^0 & \dots & \alpha_n^0 \\ \alpha_1^1 & \alpha_2^1 & \dots & \alpha_n^1 \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{k-1} & \alpha_2^{k-1} & \dots & \alpha_n^{k-1} \end{pmatrix}}_{G} \end{aligned}$$

□

Definition 4.28 (Generalized Reed-Solomon (GRS) codes)**.**

$$GRS_q(\alpha, k, v) = \{(v_1 u(\alpha_1), \dots, v_n u(\alpha_n)) \mid u(x) \in \mathbb{F}_q[x], \deg u \leq k-1\}$$

where $v = (v_1, \dots, v_n) \in (\mathbb{F}_q^*)^n$.

If $v = (1, \dots, 1)$, then

$$GRS_q(\alpha, k, v) = RS_q(\alpha, k).$$

Remark 4.29. (i) $GRS_q(\alpha, k, v)$ is an $[n, k, n-k+1]$ linear code over $\mathbb{F}_q$.

(ii) A generator matrix of $GRS_q(\alpha, k, v)$ is

$$\begin{pmatrix} v_1 & v_2 & \dots & v_n \\ v_1\alpha_1 & v_2\alpha_2 & \dots & v_n\alpha_n \\ \vdots & \vdots & \ddots & \vdots \\ v_1\alpha_1^{k-1} & v_2\alpha_2^{k-1} & \dots & v_n\alpha_n^{k-1} \end{pmatrix}.$$

(iii) $v_1, \ldots, v_n$ are called column multipliers.

(iv) The dual code of $GRS_q(\alpha, k, v)$ is $GRS_q(\alpha, n-k, v')$ for some $v' \in (\mathbb{F}_q^*)^n$.

4.8 Decoding of RS codes

The decoding problem:

Given an $RS_q(\alpha, k)$ code (where $\alpha = (\alpha_1, \ldots, \alpha_n)$ and minimum distance $d = n-k+1$) and a received word $y = (y_1, \ldots, y_n) \in \mathbb{F}_q^n$ with at most $\lfloor \frac{n-k}{2} \rfloor$ errors, we want to (efficiently) find $f(x) \in \mathbb{F}_q[x]$ with degree $\deg f \leq k-1$ s.t.

$$d\left(y, (f(\alpha_1), \ldots, f(\alpha_n))\right) \leq \left\lfloor \frac{n-k}{2} \right\rfloor.$$

The transmission through the channel can be represented as:

$$(f(\alpha_1), \ldots, f(\alpha_n)) \xrightarrow{\text{channel}} y = (y_1, \ldots, y_n)$$

Let J be the set of error locations:

$$|J| = |\{i \mid y_i \neq f(\alpha_i)\}| \leq \left\lfloor \frac{n-k}{2} \right\rfloor$$

If we know the error locations $J \subset \{1, \ldots, n\}$, $|J| \leq \lfloor \frac{n-k}{2} \rfloor$, then we can use the correct coordinates

$$\{(\alpha_i, y_i) \mid i \in \{1, \ldots, n\} \setminus J\}$$

to find $f(x)$, **using polynomial interpolation**.

Since $\deg f \leq k-1$, we need k correct coordinates to determine f.

Since $|J| \leq \lfloor \frac{n-k}{2} \rfloor$, the number of correct coordinates is

$$|\{1, \ldots, n\} \setminus J| \geq n - \left\lfloor \frac{n-k}{2} \right\rfloor \geq \frac{n+k}{2} > \deg f.$$

Let $I \subset \{1, \ldots, n\} \setminus J$, with $|I| = k$.

Then

$$f(x) = \sum_{i \in I} y_i \prod_{j \in I \setminus \{i\}} \frac{x - \alpha_j}{\alpha_i - \alpha_j}.$$

Verify: $f(\alpha_s) = y_s$ for all $s \in I$. However, we do not know the error locations.

Idea 1: Find the Error Locations

Definition 4.30 (Error-locator polynomials)**.** A polynomial $E(x) \in \mathbb{F}_q[x]$ is called an error-locator polynomial if for all $i \in \{1, \ldots, n\}$ we have

$$E(\alpha_i) = 0 \quad \text{if} \quad y_i \neq f(\alpha_i).$$

Remark 4.31. (i) The roots of $E(x)$ contains the error locations.

(ii) There exists at least one error-locator polynomials:

$$E(x) := \prod_{i: y_i \neq f(\alpha_i)} (x - \alpha_i).$$

(iii) It holds that

$$y_i E(\alpha_i) = f(\alpha_i) E(\alpha_i) \quad \text{for all } i. \tag{*}$$

$$y_i \sum_{s=0}^{\deg E} e_s \alpha_i^s = \left(\sum_{j=0}^{\deg f} f_j \alpha_i^j \right) \left(\sum_{s=0}^{\deg E} e_s \alpha_i^s \right) \quad \forall i$$

(where y_i and α_i are known, while e_s and f_j are unknown)

The system (*) is a system of n quadratic equations.

Idea 2: Convert (*) to linear equations by introducing new variables.

Define

$$N(x) := f(x)E(x)$$

Now (*) becomes

$$y_i E(\alpha_i) = N(\alpha_i)$$

$$y_i \sum_{s=0}^{\deg E} e_s \alpha_i^s = \sum_{j=0}^{\deg N} n_j \alpha_i^j \quad \forall i$$

(where y_i and α_i are known, while e_s and n_j are unknown)

If one can solve these linear equations to find $E(x), N(x)$, then $f(x)$ can be found by setting

$$f(x) = \frac{N(x)}{E(x)}.$$

However, we still need to ensure $E(x) \mid N(x)$.

So although we have a set of linear equations, the condition that $E(x) \mid N(x)$ makes it difficult to solve.

Idea 3: "Forget" about the constraint $E(x) \mid N(x)$ and solve the linear system to find candidate solutions.

Just check each candidate if it satisfies the constraint.

Remark 4.32. If the number of candidate solutions is small, then we have an efficient decoding algorithm.

The Welch-Berlekamp (WB) algorithm

Input: $n \geq k \geq 1$ and $\{(\alpha_i, y_i) \mid i = 1, \ldots, n\}$.

Output: a polynomial $f(x)$ with $\deg f \leq k-1$ or FAIL.

Steps:

1. Compute an error-locator polynomial $E(x)$ with $\deg E \leq \left\lfloor \frac{n-k}{2} \right\rfloor$ and a polynomial $N(x)$ with $\deg N \leq \left\lfloor \frac{n-k}{2} \right\rfloor + k - 1$ s.t.

$$y_i E(\alpha_i) = N(\alpha_i), \quad 1 \leq i \leq n. \tag{**}$$

2. If $E(x)$ and $N(x)$ as above do not exist or $E(x)$ does not divide $N(x)$, then output FAIL.
3. Otherwise, define $f(x) := \frac{N(x)}{E(x)}$.
4. If $d\left(y, (f(\alpha_1), \ldots, f(\alpha_n))\right) > \left\lfloor \frac{n-k}{2} \right\rfloor$ then output FAIL.
5. Otherwise, output $f(x)$.

Correctness of the WB algorithm:

Existence of a nontrivial solution.

Proposition 4.33. *If there are at most $\lfloor \frac{n-k}{2} \rfloor$ errors, then there exist $E^*(x), N^*(x)$ s.t. (**) is satisfied and $E^*(x) \mid N^*(x)$.*

Proof. Let J be the set of error locations. Then $|J| \leq \lfloor \frac{n-k}{2} \rfloor$.

Define $E^*(x) = \prod_{i \in J}(x - \alpha_i)$. Then

$$\deg E^* = |J| \leq \left\lfloor \frac{n-k}{2} \right\rfloor.$$

Define $N^*(x) = f(x)E^*(x)$. Then

$$\deg N^* = \deg E^* + \deg f \leq \left\lfloor \frac{n-k}{2} \right\rfloor + k - 1.$$

Clearly, by construction, $E^*(x) \mid N^*(x)$.

If $E^*(\alpha_i) = 0$ for some i, then

$$N^*(\alpha_i) = 0 \quad \text{and} \quad y_i E^*(\alpha_i) = 0.$$

Namely, $N^*(\alpha_i) = y_i E^*(\alpha_i)$.

If $E^*(\alpha_i) \neq 0$, then i is not an error location. Therefore, $y_i = f(\alpha_i)$ and

$$N^*(\alpha_i) = f(\alpha_i)E^*(\alpha_i) = y_i E^*(\alpha_i).$$

$\square$

Proposition 4.34 (Uniqueness of the solution)**.** *If there exists* $(N(x), E(x)) \neq (N^*(x), E^*(x))$ *that satisfies Step 1, then*

$$\frac{N(x)}{E(x)} = \frac{N^*(x)}{E^*(x)}.$$

Proof. Define:

$$R(x) = N(x)E^*(x) - N^*(x)E(x).$$

The degree of $R(x)$ is bounded by:

$$\deg R \leq \deg N + \deg E^* \leq \left\lfloor \frac{n-k}{2} \right\rfloor + k - 1 + \left\lfloor \frac{n-k}{2} \right\rfloor = n - 1.$$

From the system of equations, we have:

$$\begin{cases} N(\alpha_i) = y_i E(\alpha_i) \\ N^*(\alpha_i) = y_i E^*(\alpha_i) \end{cases} \qquad i = 1, \ldots, n.$$

Substituting these into the expression for $R(\alpha_i)$:

$$N(\alpha_i)E^*(\alpha_i) = y_i E(\alpha_i)E^*(\alpha_i) = E(\alpha_i)N^*(\alpha_i), \quad i = 1, \ldots, n.$$

So α_i is a root of $R(x)$ for all $i = 1, \ldots, n$.

Since $\deg R \leq n - 1$ but $R(x)$ has n distinct roots, it follows that $R(x) \equiv 0$. Therefore,

$$N(x)E^*(x) \equiv N^*(x)E(x),$$

which implies

$$\frac{N(x)}{E(x)} = \frac{N^*(x)}{E^*(x)}.$$

□

Remark 4.35*.* The WB algorithm is a bivariate polynomial interpolation algorithm. The idea is to determine:

$$Q(x, y) = N(x) - yE(x) \in \mathbb{F}_q[x, y] \setminus \{0\}$$

s.t.

- $Q(\alpha_i, y_i) = 0, \quad i = 1, \ldots, n;$
- $\deg N \leq \left\lfloor \frac{n-k}{2} \right\rfloor + k - 1;$
- $\deg E \leq \left\lfloor \frac{n-k}{2} \right\rfloor.$

Chapter 5

Asymptotic Analysis

5.1 Asymptotic Notation

Let f, g be functions from $\mathbb{Z}^+$ to $\mathbb{R}^+$.

Big-O notation:

We say $f(n)$ is $O(g(n))$ iff there exists $K > 0$ s.t.

$$\limsup_{n\to\infty} \frac{f(n)}{g(n)} < K.$$

E.g., $f(n) = 24n^2 + 100n$

- $f(n) = O(n^2)$
- $f(n) = O(n^3)$
- $f(n) \neq O(n)$

Big-Omega notation:

We say $f(n)$ is $\Omega(g(n))$ iff there exists $K > 0$ s.t.

$$\liminf_{n\to\infty} \frac{f(n)}{g(n)} > K.$$

E.g., $f(n) = 24n^2 + 100n$

- $f(n) = \Omega(n^2)$
- $f(n) \neq \Omega(n^3)$

$$\lim_{n\to\infty} \frac{24n^2 + 100n}{n^3} = \lim_{n\to\infty} \left(\frac{24}{n} + \frac{100}{n^2}\right) = 0$$

- $f(n) = \Omega(n^{3/2})$

Theta notation:

We say $f(n)$ is $\Theta(g(n))$ if

$$f(n) = O(g(n)) \text{ and } f(n) = \Omega(g(n)).$$

E.g., $f(n) = 24n^2 + 100n = \Theta(n^2)$

Little-oh notation:

We say $f(n)$ is $o(g(n))$ iff

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0.$$

E.g.,

- $24n^2 + 100n = o(n^3)$
- $24n^2 + 100n \neq o(n^2)$
- $f(n) = \frac{1}{n} = o(1)$

Little-omega notation: We say $f(n)$ is $\omega(g(n))$ iff

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \infty$$

E.g.,

- $24n^2 + 100n = \omega(n)$
- $24n^2 + 100n \neq \omega(n^2)$

Running time analysis of WB:

1. **Step 1:** Need to solve n linear equations with $O(n)$ variables. This requires $O(n^3)$ time.
2. **Steps 2, 3:** Need to check $E(x) \mid N(x)$ or calculate $\frac{N(x)}{E(x)}$. The long division algorithm can be done in time $O(n^2)$.

5.2 Asymptotic Bounds on Code Parameters

Let C be an (n, M, d) code over an alphabet of size q.

The relative (minimum) distance of C is the ratio

$$\delta = \frac{d}{n}$$

For a fixed q, relations between δ and $R = \frac{\log_q M}{n}$ as $n \to \infty$ are called asymptotic bounds on the parameters.

Definition 5.1 (Code families)**.** Let $\{n_i\}_{i\geq 0}$ be an increasing sequence of block lengths and suppose there exist $\{k_i\}_{i\geq 1}, \{d_i\}_{i\geq 1}$ s.t. for all $i \geq 1$, there exists an (n_i, k_i, d_i) code C_i over an alphabet of size q. The sequence $\mathcal{C} = \{C_i\}_{i\geq 1}$ is a family of (linear) codes.

The rate of $\mathcal{C}$ is

$$R(\mathcal{C}) := \lim_{i\to\infty}\left\{\frac{k_i}{n_i}\right\} \text{ if the limit exists.}$$

The relative distance of $\mathcal{C}$ is

$$\delta(\mathcal{C}) = \lim_{i\to\infty}\left\{\frac{d_i}{n_i}\right\} \text{ if the limit exists.}$$

Example 5.2. The Hamming code family:

$\mathcal{C}_H = \{Ham(r,q) \mid r = 2,3,\dots\}$
$Ham(r,q)$ is a $\left(\frac{q^r-1}{q-1}, \frac{q^r-1}{q-1} - r, 3\right)$ linear code over $\mathbb{F}_q$.
The rate of the Hamming code family is:

$$R(\mathcal{C}_H) = \lim_{r\to\infty}\frac{k_r}{n_r} = \lim_{r\to\infty}\frac{n_r - r}{n_r} = 1.$$

The relative distance is:

$$\delta(\mathcal{C}_H) = \lim_{r\to\infty}\frac{3}{n_r} = 0.$$

Example 5.3. The family of repetition codes satisfies $R \to 0, \delta \to 1$.

The Singleton bound:

For any (n, M, d) code over an alphabet of size q, we have:

$$d \leq n - \log_q M + 1.$$

Dividing by n, we obtain:

$$\frac{d}{n} \leq 1 - \frac{\log_q M}{n} + \frac{1}{n}.$$

The asymptotic version of the Singleton bound:

$$\delta \leq 1 - R + o(1)$$

5.3 Plotkin Bound

Lemma 5.4 (Cauchy–Schwarz inequality)**.** *Let* $a = (a_1, \dots, a_m) \in \mathbb{R}^m$*,* $b = (b_1, \dots, b_m) \in \mathbb{R}^m$*. Then*

$$\left(\sum_{r=1}^{m} a_r b_r\right)^2 \leq \left(\sum_{i=1}^{m} a_i^2\right)\left(\sum_{j=1}^{m} b_j^2\right)$$

Sketch Proof.

$$\langle a, b\rangle^2 \leq \|a\|_2^2 \|b\|_2^2.$$

□

Theorem 5.5 (Plotkin bound)**.** *Let q, n, d be integers s.t. $q > 1$ and $rn < d$ where $r := 1 - q^{-1}$. Then*

$$A_q(n,d) \le \left\lfloor \frac{d}{d - rn} \right\rfloor .$$

Proof. Let C be an (n, M, d) code over an alphabet Q of size q, where $M > 1$. Let $r := 1 - q^{-1}$. Define the total Hamming distance over all ordered pairs of codewords as:

$$T := \sum_{c \in C} \sum_{c' \in C} d(c, c')$$

Since the minimum distance of the code is d, we know that $d(c, c') \ge d$ if $c \ne c'$, and $d(c, c') = 0$ if $c = c'$. There are $M(M-1)$ pairs of distinct codewords, so:

$$T = \sum_{c \in C} \sum_{c' \ne c} d(c, c') \ge M(M-1)d$$

Alternatively, we can calculate T by summing the distances coordinate-wise. Let $c = (c_1, \dots, c_n)$ and $c' = (c'_1, \dots, c'_n)$. Then:

$$T = \sum_{c \in C} \sum_{c' \in C} \sum_{i=1}^{n} d(c_i, c'_i) = \sum_{i=1}^{n} \left(\sum_{c \in C} \sum_{c' \in C} d(c_i, c'_i) \right)$$

For a fixed coordinate i and each symbol $a \in Q$, let $n_{i,a}$ be the number of codewords in C whose i-th coordinate is a. Then $\sum_{a \in Q} n_{i,a} = M$. For a fixed i, a pair of codewords (c, c') contributes 1 to the distance if $c_i \ne c'_i$. Thus:

$$\sum_{c \in C} \sum_{c' \in C} d(c_i, c'_i) = \sum_{a \in Q} n_{i,a}(M - n_{i,a}) = M \sum_{a \in Q} n_{i,a} - \sum_{a \in Q} n_{i,a}^2 = M^2 - \sum_{a \in Q} n_{i,a}^2$$

Summing over all coordinates i:

$$T = \sum_{i=1}^{n} \left(M^2 - \sum_{a \in Q} n_{i,a}^2 \right) = nM^2 - \sum_{i=1}^{n} \sum_{a \in Q} n_{i,a}^2$$

By the Cauchy–Schwarz inequality, using vectors $x = (1, 1, \dots, 1)$ and $y = (n_{i,a})_{a \in Q}$ in $\mathbb{R}^q$:

$$\langle x, y \rangle^2 \le \|x\|_2^2 \|y\|_2^2 \implies \left(\sum_{a \in Q} n_{i,a} \right)^2 \le q \sum_{a \in Q} n_{i,a}^2 \implies M^2 \le q \sum_{a \in Q} n_{i,a}^2$$

This implies $\sum_{a \in Q} n_{i,a}^2 \ge \frac{M^2}{q}$. Substituting this into the expression for T:

$$T \le nM^2 - n\frac{M^2}{q} = nM^2 \left(1 - \frac{1}{q} \right) = nM^2 r$$

Combining the lower and upper bounds for T:

$$M(M-1)d \leq T \leq nM^2 r$$

Dividing by M (since $M > 1$):

$$\begin{aligned}(M-1)d &\leq nMr\\ Md - d &\leq nMr\\ d &\geq Md - nMr = M(d - nr)\\ M &\leq \frac{d}{d - nr}\end{aligned}$$

Since M must be an integer, we have $A_q(n,d) \leq \left\lfloor \frac{d}{d-nr} \right\rfloor$. □

Since we know $\delta \leq 1 - R + o(1)$

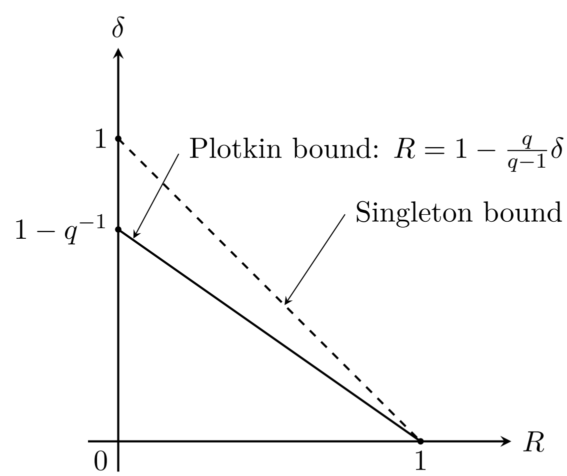
GRS codes achieve the Singleton bound but require $n \leq q$. So as n increases, q must also increase.

From the Plotkin bound, for $\delta > 1 - q^{-1}$, we have

$$M \leq \frac{\delta}{\delta - r} \leftarrow \text{a constant}$$

$$\begin{aligned}\Rightarrow R = \frac{\log_q M}{n} &\leq \frac{\log_q \frac{\delta}{\delta - r}}{n} \to 0 \quad \text{as } n \to \infty\\ R &= o(1).\end{aligned}$$

Hence we could draw the picture below:



Corollary 5.6. *Let $\mathcal{C}$ be a family of codes over an alphabet Q of size q with relative distance $0 \leq \delta \leq 1 - q^{-1}$ and rate R. Then*

$$R \leq 1 - \frac{q}{q-1}\delta + o(1).$$

Proof. Let $n' = \left\lfloor \frac{qd}{q-1} \right\rfloor - 1$. For every $x \in Q^{n-n'}$, define the sub-code:

$$C_x = \{(c_{n-n'+1}, \ldots, c_n) \mid (c_1, \ldots, c_n) \in C, (c_1, \ldots, c_{n-n'}) = x\}$$

Then C_x is a q-ary code of length n'. Moreover,

$$d(C_x) = \min_{\substack{a,b \in C_x \\ a \neq b}} d(a,b) \geq \min_{\substack{c,c' \in C \\ c \neq c'}} d(c,c') = d(C) = d > (1 - q^{-1})n'$$

Since $n' < \frac{qd}{q-1}$, we have $(1 - q^{-1})n' < d$. By the Plotkin bound:

$$\begin{aligned} |C_x| &\leq \left\lfloor \frac{d(C_x)}{d(C_x) - (1 - q^{-1})n'} \right\rfloor \\ &\leq \frac{d}{d - (1 - q^{-1})n'} \\ &= \frac{qd}{qd - (q-1)n'} \leq qd \end{aligned}$$

Observe that $\{C_x \mid x \in Q^{n-n'}\}$ is a "partition" of C.

$$\begin{aligned} |C| &= \sum_{x \in Q^{n-n'}} |C_x| \\ &\leq \sum_{x \in Q^{n-n'}} qd \\ &= q^{n-n'} \cdot qd \\ &= q^{n-n'+1} \cdot d \\ &= q^{n-n'+1+\log_q d} \\ &= q^{n\left(1 - \frac{n'+1}{n} + \frac{\log_q d}{n}\right)} \\ &= q^{n\left(1 - \frac{1}{n}\left\lfloor \frac{qd}{q-1} \right\rfloor + \frac{2+\log_q d}{n}\right)} \\ &\leq q^{n\left(1 - \delta\frac{q}{q-1} + o(1)\right)} \end{aligned}$$

Hence,

$$\Rightarrow R \leq 1 - \delta\frac{q}{q-1} + o(1).$$

□

Remark 5.7. The corollary implies that for any $q \geq 2$, there does not exist a q-ary code family that meets the Singleton bound, i.e., we always have $R < 1 - \delta$ for fixed q.

5.4 Asymptotic Version of the Hamming and GV Bounds

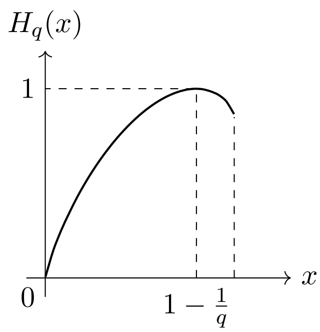
$$V_q(n,t) = \sum_{i=0}^{t} \binom{n}{i} (q-1)^i$$

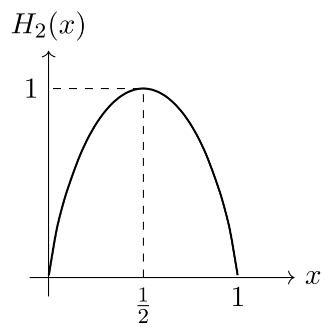
We want to find upper and lower bounds of $V_q(n,t)$ in "closed form" (no summation, binomial coefficient, etc.) in order to derive the asymptotic version of the Hamming bound and the GV bound.

Define the q-ary entropy function $H_q : [0,1] \to [0,1]$ given by

$$H_q(x) = -x \log_q x - (1-x)\log_q(1-x) + x\log_q(q-1),$$

where $q \geq 2$ is an integer.





Lemma 5.8. *For $0 \leq \frac{t}{n} \leq 1 - q^{-1}$, let $p = \frac{t}{n}$. The volume of the q-ary Hamming ball $V_q(n,t)$ satisfies:*

$$V_q(n,t) \leq q^{nH_q(p)}$$

This is equivalent to showing $q^{-nH_q(p)} V_q(n,t) \leq 1$.

Proof. First, we express $q^{-nH_q(p)}$ in terms of p. By the definition of the q-ary entropy function:

$$\begin{aligned} q^{-nH_q(p)} &= q^{-n\left(-p\log_q p - (1-p)\log_q(1-p) + p\log_q(q-1)\right)} \\ &= q^{np\log_q p + n(1-p)\log_q(1-p) - np\log_q(q-1)} \\ &= p^{np}(1-p)^{n(1-p)}(q-1)^{-np} \end{aligned}$$

Next, consider the identity $1 = (p + (1-p))^n$. Expanding this using the binomial theorem:

$$\begin{aligned} 1 = \sum_{i=0}^{n} \binom{n}{i} p^i (1-p)^{n-i} &\geq \sum_{i=0}^{pn} \binom{n}{i} p^i (1-p)^{n-i} \\ &= \sum_{i=0}^{pn} \binom{n}{i} (q-1)^i (1-p)^n \left(\frac{p}{(q-1)(1-p)}\right)^i \end{aligned}$$

Given the condition $p \le 1 - q^{-1}$, we can show that $\frac{p}{(q-1)(1-p)} \le 1$. For $0 \le i \le pn$, the term $\left(\frac{p}{(q-1)(1-p)}\right)^i$ is a non-increasing function of i, therefore:

$$\left(\frac{p}{(q-1)(1-p)}\right)^i \ge \left(\frac{p}{(q-1)(1-p)}\right)^{pn}$$

Substituting this back into our inequality:

$$\begin{aligned} 1 &\ge (1-p)^n \left(\frac{p}{(q-1)(1-p)}\right)^{pn} \sum_{i=0}^{pn} \binom{n}{i} (q-1)^i \\ &= (1-p)^n \frac{p^{np}}{(q-1)^{np}(1-p)^{np}} V_q(n,t) \\ &= p^{np}(1-p)^{n(1-p)}(q-1)^{-np} V_q(n,t) \\ &= q^{-nH_q(p)} V_q(n,t) \end{aligned}$$

Thus, $V_q(n,t) \le q^{nH_q(p)}$, which completes the proof. □

Lemma 5.9. *For integers $0 \le t \le n$, the volume of the q-ary Hamming ball satisfies:*

$$V_q(n,t) \ge \frac{1}{n+1} q^{nH_q(\frac{t}{n})} = q^{nH_q(\frac{t}{n}) - o(n)}$$

Proof. Let $p := \frac{t}{n}$. We consider the ratio of the volume to the exponential term:

$$q^{-nH_q(p)} V_q(n,t) \ge q^{-nH_q(p)} \binom{n}{t} (q-1)^t$$

Using the identity derived previously, $q^{-nH_q(p)} = \left(\frac{p}{q-1}\right)^{pn} (1-p)^{n(1-p)}$, we substitute it into the right-hand side:

$$\begin{aligned} q^{-nH_q(p)} \binom{n}{t} (q-1)^t &= \left[\left(\frac{p}{q-1}\right)^{pn} (1-p)^{n(1-p)}\right] \binom{n}{np} (q-1)^{np} \\ &= \binom{n}{np} p^{np} (1-p)^{n-np} \\ &= \binom{n}{t} p^t (1-p)^{n-t} \end{aligned}$$

Define $A_i = \binom{n}{i} p^i (1-p)^{n-i}$. From the above calculation, the term we are interested in is exactly A_t. To show A_t is the maximum term in the sequence $\{A_i\}$, consider the ratio:

$$\frac{A_{i+1}}{A_i} = \frac{\binom{n}{i+1} p^{i+1}(1-p)^{n-i-1}}{\binom{n}{i} p^i (1-p)^{n-i}} = \frac{n-i}{i+1} \cdot \frac{p}{1-p} = \frac{n-i}{i+1} \cdot \frac{t}{n-t}$$

The condition $\frac{A_{i+1}}{A_i} < 1$ holds if and only if $i \ge t$. So $A_0 \le \cdots \le A_{t-1} \le A_t \ge A_{t+1} \ge \cdots \ge A_n$.

By the binomial theorem, the sum of all A_i is 1:

$$1 = (p + (1-p))^n = \sum_{i=0}^{n} A_i \leq \sum_{i=0}^{n} A_t = (n+1)A_t$$

This implies $A_t \geq \frac{1}{n+1}$. Combining this result with our initial inequality:

$$q^{-nH_q(p)} V_q(n,t) \geq A_t \geq \frac{1}{n+1}$$

Therefore,

$$V_q(n,t) \geq \frac{1}{n+1} q^{nH_q(\frac{t}{n})}$$

Moreover,

$$\begin{aligned}\frac{1}{n+1} q^{nH_q(\frac{t}{n})} &= q^{-\log_q(n+1)} \cdot q^{nH_q(\frac{t}{n})} \\ &= q^{-o(n)} \cdot q^{nH_q(\frac{t}{n})} \\ &= q^{nH_q(\frac{t}{n}) - o(n)}\end{aligned}$$

Asymptotically. we have the following conclusion:

$$q^{nH_q(\frac{t}{n}) - o(n)} \leq V_q(n,t) \leq q^{nH_q(\frac{t}{n})}$$

□

Theorem 5.10 (Asymptotic version of Hamming bound)**.** *For every (n, M, d) code over an alphabet of size q,*

$$R \leq 1 - H_q\left(\frac{\delta}{2}\right) + o(1).$$

Proof. Recall $R = \frac{\log_q M}{n}$ and $\delta = \frac{d}{n}$. Define $t = \left\lfloor \frac{d-1}{2} \right\rfloor$.

First, we establish the asymptotic behavior of the volume $V_q(n,t)$. Note the bounds on $\frac{t}{n}$:

$$\begin{aligned}\frac{1}{2} \geq \frac{t}{n} &= \frac{1}{n} \left\lfloor \frac{d-1}{2} \right\rfloor \\ &\geq \frac{1}{n} \cdot \frac{d-2}{2} = \frac{\delta}{2} - \frac{1}{n}\end{aligned}$$

Since we know $V_q(n,t) \geq \frac{1}{n+1} \cdot q^{nH_q\left(\frac{t}{n}\right)}$, we can substitute the lower bound for $\frac{t}{n}$:

$$\begin{aligned}V_q(n,t) &\geq \frac{1}{n+1} \cdot q^{nH_q\left(\frac{\delta}{2} - \frac{1}{n}\right)} \\ &= q^{n\left(H_q\left(\frac{\delta}{2} - o(1)\right) - o(1)\right)} \\ &= q^{n\left(H_q\left(\frac{\delta}{2}\right) - o(1)\right)}\end{aligned}$$

Now, by the standard Hamming bound, we have $M \leq \frac{q^n}{V_q(n,t)}$. Substituting the asymptotic lower bound for $V_q(n,t)$ gives:

$$q^{nR} = M \leq \frac{q^n}{q^{n\left(H_q\left(\frac{\delta}{2}\right)-o(1)\right)}} = q^{n\left(1-H_q\left(\frac{\delta}{2}\right)+o(1)\right)}$$

Taking the logarithm base q and dividing by n yields the final result:

$$R \leq 1 - H_q\left(\frac{\delta}{2}\right) + o(1).$$

□

Theorem 5.11 (Asymptotic version of the GV bound). *Let n, nR be positive integers and let $\delta \in [0, 1-q^{-1}]$ be such that*

$$R \leq 1 - H_q(\delta).$$

Then there exists an $[n, nR, \geq \delta n]$ linear code over $\mathbb{F}_q$.

Proof. By the GV bound, if

$$V_q(n-1, \lceil \delta n \rceil - 2) \leq q^{n(1-R)}$$

then there exists an $[n, nR, \geq \delta n]$ linear code over $\mathbb{F}_q$.

Notice that the volume term can be strictly bounded:

$$V_q(n-1, \lceil \delta n \rceil - 2) < V_q(n, \lfloor \delta n \rfloor) \leq q^{nH_q(\delta)}$$

So, if

$$q^{nH_q(\delta)} \leq q^{n(1-R)}, \quad \text{i.e.,} \quad R \leq 1 - H_q(\delta),$$

then there exists an $[n, nR, \geq \delta n]$ linear code over $\mathbb{F}_q$. □

Remark 5.12. Since $H_q(\delta) < 1$ whenever $\delta < 1 - q^{-1}$, the GV bound implies for $\delta < 1 - q^{-1}$, there exists a family of codes over an alphabet of size q such that $R > 0, \delta > 0$. (Asymptotically good codes)

Chapter 6

Construct New Codes from Existing Codes

6.1 Puncturing Codes

Suppose C is an $[n, k, d]$ linear code over $\mathbb{F}_q$.

Definition 6.1 (Puncturing codes:remove a coordinate from C)**.** By puncturing C on the i-th coordinate, we mean the i-th coordinate in each codeword of C is deleted.

Theorem 6.2. *Let C_i^* be the code C punctured on the i-th coordinate. If $d \geq 2$, then C_i^* is an $[n-1, k, d^*]$ linear code with $d - 1 \leq d^* \leq d$.*

Proof. If no codewords of weight d in C whose i-th coordinate is nonzero, then $d^* = d$. Otherwise, $d^* = d - 1$. □

Remark 6.3. When $d = 1$, if no codewords of weight 1 in C whose i-th coordinate is nonzero, $d^* = d = 1$. Otherwise, if $k > 1$, then C^* is an $[n-1, k-1, d^* \geq 1]$ linear code.

Example 6.4. Let C be given by

$$G = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

over $\mathbb{F}_2$. So C is a $[5, 2, 2]$ linear code.

C_1^* has the following generator matrix:

$$G_1^* = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

So C_1^* is $[4, 2, 1]$.

6.2 Extending Codes

Definition 6.5 (Add a coordinate to C)**.** The extended code of C is defined to be

$$\widehat{C} = \left\{ (c_1, \ldots, c_n, c_{n+1}) \mid (c_1, \ldots, c_n) \in C, \sum_{i=1}^{n+1} c_i = 0 \right\}.$$

Theorem 6.6. *The extended code $\widehat{C}$ is an $[n+1, k, \widehat{d}]$ linear code with $d \leq \widehat{d} \leq d+1$. If H is a parity-check matrix of C (size $(n-k) \times n$), then a parity-check matrix of $\widehat{C}$ can be given by*

$$\widehat{H} = \left(\begin{array}{ccc|c} & & & 0 \\ & H & & \vdots \\ & & & 0 \\ \hline 1 & \ldots & 1 & 1 \end{array} \right)_{(n-k+1)\times(n+1)}$$

Proof. $d(\widehat{C})$ can be increased by at most 1 from $d(C)$.

Let $c = (c_1, \ldots, c_n) \in C$. Then $Hc^T = 0$.

Define $\widehat{c} = (c_1, \ldots, c_n, -\sum c_i)$. Then $\widehat{c} \in \widehat{C}$.

Moreover,

$$\widehat{H}\widehat{c}^T = \left(\begin{array}{ccc|c} & & & 0 \\ & H & & \vdots \\ & & & 0 \\ \hline 1 & \ldots & 1 & 1 \end{array} \right) \begin{pmatrix} c_1 \\ \vdots \\ c_n \\ -\sum c_i \end{pmatrix} = 0$$

Hence, $\{x \in \mathbb{F}_q^{n+1} \mid \widehat{H}x^T = 0\} \subseteq \widehat{C}$

On the other hand, for $x = (x_1, \ldots, x_{n+1}) \in \mathbb{F}_q^{n+1}$ with $\widehat{H}x^T = 0$, we have

$$H(x_1, \ldots, x_n)^T = 0.$$

So $(x_1, \ldots, x_n) \in C \implies x_{n+1} = -\sum_{i=1}^n x_i$. Therefore, $x \in \widehat{C}$.

$$\{x \in \mathbb{F}_q^{n+1} \mid \widehat{H}x^T = 0\} \subseteq \widehat{C}$$

Then, $\{x \in \mathbb{F}_q^{n+1} \mid \widehat{H}x^T = 0\} = \widehat{C}$. □

Example 6.7.

$$C = \{000, 110, 011, 101\} \text{ over } \mathbb{F}_2, \quad d(C) = 2$$

$$\widehat{C} = \{0000, 1100, 0110, 1010\} \text{ over } \mathbb{F}_2, \quad d(\widehat{C}) = 2$$

Remark 6.8.
- Augmenting codes (add codewords to C) $\longleftrightarrow$ Expurgating codes (remove codewords from C)
- Extending codes $\longleftrightarrow$ Puncturing codes

- A linear subcode of C is an example of expurgated codes.

Definition 6.9. Let T be any set of t coordinates. Define

$$C(T) = \{(c_1, \dots, c_n) \mid (c_1, \dots, c_n) \in C, c_i = 0 \text{ for } i \in T\}.$$

Theorem 6.10. *If $|T| \le k-1$, then there exists an $[n, k-t]$ linear code $C(T)$ over $\mathbb{F}_q$.*

Proof. Assume (without loss of generality) that C has a generator matrix in standard form, i.e.,

$$G = (I_k \mid X).$$

Let $T = \{1, \dots, t\}$, $t \le k-1$. Write G in block form as:

$$G = \left(\begin{array}{cc|c} I_t & 0 & \multirow{2}{*}{X} \\ 0 & I_{k-t} & \end{array} \right)$$

Define $G(T)$ to be formed by the last $k-t$ rows of G. Then $G(T)$ generates an $[n, k-t]$ linear code over $\mathbb{F}_q$, whose codewords have zeros on T. □

Remark 6.11.
- $d(C(T)) \ge d(C)$
- Extending + augmenting ⇒ Lengthening
- Expurgating + puncturing ⇒ Shortening

Definition 6.12. Puncturing $C(T)$ on T gives a code of length $n - |T|$ over $\mathbb{F}_q$, called the code shortened on T and denoted by C_T.

Theorem 6.13. *If $|T| \le k-1$, then there exists an $[n - |T|, k - |T|]$ linear code C_T over $\mathbb{F}_q$.*

Proof. Without loss of generality, assume C has a parity-check matrix in the following form:

$$H = (I_{n-k} \mid X)$$

Delete the last $|T|$ columns of H to obtain

$$H_T = (I_{n-k} \mid Y),$$

where Y is formed by the first $k - |T|$ columns of X.

H_T has full rank. Any $d-1$ columns of H_T are linearly independent.

H_T defines a linear code

$$\widetilde{C} = \{x \in \mathbb{F}_q^{n-|T|} \mid H_T x^T = 0\}$$

of dimension $n - |T| - (n-k) = k - |T|$, and

$$d(\widetilde{C}) \ge d.$$

Claim : $\widetilde{C} = C_T$

For $\tilde{c} \in \widetilde{C}$, we have $H(\tilde{c}, 0, \ldots, 0)^T = 0$, which gives

$$(\tilde{c}, 0, \ldots, 0) \in C \implies \widetilde{C} \subseteq C_T.$$

At the same time, if $c \in C$ s.t.

$$c = (x, \underbrace{0, \ldots, 0}_{|T|}),$$

then $Hc^T = 0 \implies H_T x^T = 0$.

Thus, $x \in \widetilde{C}$ and $C_T \subseteq \widetilde{C}$. □

6.3 Combining Codes

Theorem 6.14 (Direct Sum). *Let C_i be an $[n_i, k_i, d_i]$ linear code over $\mathbb{F}_q$ for $i = 1, 2$. Then the direct sum of C_1 and C_2 defined by*

$$C_1 \oplus C_2 = \{(c_1, c_2) \mid c_1 \in C_1, c_2 \in C_2\}$$

is an $[n_1 + n_2, k_1 + k_2, \min\{d_1, d_2\}]$ linear code over $\mathbb{F}_q$.

Proof. Length and linearity are easy to check.

$$\begin{aligned}\dim_{\mathbb{F}_q} C_1 \oplus C_2 &= \log_q |C_1 \oplus C_2| \\ &= \log_q(|C_1| \cdot |C_2|) \\ &= \log_q(q^{k_1} \cdot q^{k_2}) \\ &= k_1 + k_2.\end{aligned}$$

For any $c \in C_1 \oplus C_2$ with $c = (c_1, c_2)$, $c_1 \in C_1$, $c_2 \in C_2$, we have

$$\begin{aligned}wt(c) &= wt((c_1, c_2)) \\ &= wt(c_1) + wt(c_2).\end{aligned}$$

If $c \neq 0$, then $c_1 \neq 0$ or $c_2 \neq 0$.

(i) If $c_1 \neq 0$, then

$$wt(c) = wt(c_1) + \underbrace{wt(c_2)}_{\geq 0} \geq wt(c_1) \geq d_1.$$

(ii) If $c_2 \neq 0$, then

$$wt(c) = wt(c_1) + wt(c_2) \geq wt(c_2) \geq d_2.$$

Then if $c \neq 0$, $wt(c) \geq \min\{d_1, d_2\}$.

Moreover, choosing $c_1 \in C_1$ such that $wt(c_1) = d_1$, we have $(c_1, 0) \in C_1 \oplus C_2$ which implies $wt((c_1, 0)) = d_1$. Similarly, there exists $c_2 \in C_2$ such that $wt((0, c_2)) = d_2$. Thus, $d(C_1 \oplus C_2) = \min\{d_1, d_2\}$. $\square$

Remark 6.15. Let G_i be a generator matrix for C_i, $i = 1, 2$. Then

$$G = \begin{pmatrix} G_1 & O \\ O & G_2 \end{pmatrix}_{(k_1+k_2)\times(n_1+n_2)}$$

is a generator matrix for $C_1 \oplus C_2$.

Indeed, for $u \in \mathbb{F}_q^{k_1+k_2}$ and $u = (u_1, u_2)$ where $u_1 \in \mathbb{F}_q^{k_1}, u_2 \in \mathbb{F}_q^{k_2}$, we have:

$$uG = (u_1G_1, u_2G_2).$$

Since $u_1G_1 \in C_1$ and $u_2G_2 \in C_2$, it follows that $(u_1G_1, u_2G_2) \in C_1 \oplus C_2$.

Theorem 6.16 ($(u, u+v)$-Construction)**.** *Let C_i be an $[n, k_i, d_i]$ linear code over $\mathbb{F}_q$ for $i = 1, 2$. Then the code C defined by*

$$C = \{(u, u+v) \mid u \in C_1, v \in C_2\}$$

is a $[2n, k_1 + k_2, \min\{2d_1, d_2\}]$ linear code over $\mathbb{F}_q$.

Proof. Length and linearity are easy to check.

The mapping $C_1 \oplus C_2 \to C$ defined by $(c_1, c_2) \mapsto (c_1, c_1 + c_2)$ is a bijection. Thus,

$$q^{k_1+k_2} = |C_1 \oplus C_2| = |C| \implies \dim_{\mathbb{F}_q} C = k_1 + k_2.$$

For any $c \in C \setminus \{0\}$ with $c = (c_1, c_1 + c_2)$, where $c_1 \in C_1, c_2 \in C_2$, we have $c_1 \neq 0$ or $c_1 + c_2 \neq 0$.

(i) If $c_2 = 0$, then

$$\left.\begin{array}{r} c_1 \neq 0 \\ c_1 + c_2 \neq 0 \end{array}\right\} \implies c_1 \neq 0.$$

Then:

$$\begin{aligned} wt(c) &= wt(c_1) + wt(c_1 + c_2) \\ &= wt(c_1) + wt(c_1) \\ &= 2wt(c_1) \\ &\geq 2d_1. \end{aligned}$$

(ii) If $c_2 \neq 0$, then

$$\begin{aligned} wt(c) &= wt(c_1) + wt(c_1 + c_2) \\ &\geq wt(c_1) + (wt(c_2) - wt(c_1)) \\ &= wt(c_2) \\ &\geq d_2. \end{aligned}$$

Therefore, for any $c \in C \setminus \{0\}$, $wt(c) \geq \min\{2d_1, d_2\}$.

Let $x \in C_1, y \in C_2$ be such that $wt(x) = d_1$ and $wt(y) = d_2$. Then $(x, x+0)$ and $(0, 0+y)$ are both in C. Moreover,

$$\begin{aligned} wt(x, x+0) &= 2wt(x) = 2d_1, \\ wt(0, 0+y) &= wt(y) = d_2. \end{aligned}$$

Thus, $d(C) = \min\{2d_1, d_2\}$. □

Remark 6.17. Assume G_i is a generator matrix for C_i, $i = 1, 2$. Then

$$G = \begin{pmatrix} G_1 & G_1 \\ 0 & G_2 \end{pmatrix}$$

is a generator matrix for C.

Indeed, for any $u = (u_1, u_2) \in \mathbb{F}_q^{k_1+k_2}$ where $u_1 \in \mathbb{F}_q^{k_1}$ and $u_2 \in \mathbb{F}_q^{k_2}$:

$$\begin{aligned} uG &= u_1(G_1, G_1) + u_2(0, G_2) \\ &= (u_1 G_1, u_1 G_1 + u_2 G_2) \in C \end{aligned}$$

Corollary 6.18. *Let A be a binary $[n, k, d]$ linear code. Define*

$$C = \{(c, c) \mid c \in A\} \cup \{(c, c+1) \mid c \in A\},$$

where $1 = (1, \ldots, 1)$ is the all-one vector of length n.

Then C is a binary $[2n, k+1, \min\{2d, n\}]$ linear code.

Proof. The code C can be viewed as the $(u, u+v)$-construction of A and the binary repetition code of length n.

Let:

- $C_1 = A$ (an $[n, k, d]$ linear code)
- $C_2 = \{\underbrace{(0, \ldots, 0)}_{n}, \underbrace{(1, \ldots, 1)}_{n}\}$ (the binary $[n, 1, n]$ repetition code)

Recall the parameters for the $(u, u+v)$-construction:

- Length: $2n$
- Dimension: $k_1 + k_2 = k + 1$
- Minimum distance: $\min\{2d_1, d_2\} = \min\{2d, n\}$

Thus, C is a binary $[2n, k+1, \min\{2d, n\}]$ linear code. □

6.4 Reed-Muller(RM) Codes

Definition 6.19. The (first-order) Reed-Muller codes $RM(1,m)$ are binary codes defined recursively for all integers $m \geq 1$ as follows:

(i) $RM(1,1) = \mathbb{F}_2^2 = \{(0,0),(0,1),(1,0),(1,1)\}$

(ii) For $m \geq 2$,

$$RM(1,m) = \{(u,u) \mid u \in RM(1,m-1)\} \cup \{(u,u+1) \mid u \in RM(1,m-1)\}.$$

Remark 6.20. Let C_n be the binary repetition code of length n. Then for $m \geq 2$,

$$RM(1,m) = \{(u,u+v) \mid u \in RM(1,m-1), v \in C_{2^m}\}.$$

Example 6.21.

$$\begin{aligned} RM(1,2) &= \{(u,u) \mid u \in \mathbb{F}_2^2\} \cup \{(u,u+1) \mid u \in \mathbb{F}_2^2\} \\ &= \{(0000),(0101),(1010),(1111)\} \cup \{(0011),(0110),(1001),(1100)\}. \end{aligned}$$

Proposition 6.22. *For $m \geq 1$, $RM(1,m)$ is a binary $[2^m, m+1, 2^{m-1}]$ linear code, in which every codeword except 0 and 1 has weight 2^{m-1}.*

Proof. By induction on m.

Base case $m = 1$: $RM(1,1) = \mathbb{F}_2^2$ is a $[2,2,1]$ code. Substituting $m = 1$ into the parameters: $[2^1, 1+1, 2^{1-1}] = [2,2,1]$. This matches.

Induction Hypothesis: Assume $RM(1,m)$ is a binary $[2^m, m+1, 2^{m-1}]$ code.

Induction Step: Consider $RM(1,m+1)$. Based on the recursive definition, $RM(1,m+1)$ is the $(u,u+v)$-construction of $RM(1,m)$ and the binary repetition code of length 2^m (denoted C_2). The codewords are of the form:

$$(u,u) \quad \text{and} \quad (u,u+1), \quad \text{where } u \in RM(1,m).$$

The parameters of $RM(1,m+1)$ are:

- Length: $2 \cdot 2^m = 2^{m+1}$
- Dimension: $(m+1)+1 = m+1+1$

- Minimum distance: $\min\{2 \cdot 2^{m-1}, 2^m\} = 2^m$

Thus, $RM(1, m+1)$ is a $[2^{m+1}, m+2, 2^m]$ binary linear code. $\square$

6.5 Cyclic codes

Definition 6.23 (Cyclic set)**.** A subset $S \subseteq \mathbb{F}_q^n$ is cyclic if the following holds:

$$(a_0, \dots, a_{n-1}) \in S \Rightarrow (a_{n-1}, a_0, \dots, a_{n-2}) \in S.$$

Definition 6.24. A linear code C is called cyclic if C is a cyclic set.

Example 6.25. The parity code $\{000, 110, 101, 011\}$ is a cyclic code.

Definition 6.26. To have an algebraic description of cyclic codes, let us define the mapping:

$$\pi : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q[x]/(x^n - 1)$$

$$(a_0, \dots, a_{n-1}) \longmapsto a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}.$$

Note that:

- π is $\mathbb{F}_q$-linear.
- π is bijective.

Definition 6.27 (Ideal)**.** Let R be a ring. A nonempty subset $I \subseteq R$ is called an **ideal** if:

(i) For all $a, b \in I$, $\begin{cases} a + b \in I \\ a - b \in I \end{cases}$

(ii) For all $a \in I, r \in R$, $\begin{cases} a \cdot r \in I \\ r \cdot a \in I \end{cases}$

Definition 6.28 (Principal ideals)**.** An ideal of a ring R is called a **principal ideal** if there exists an element $g \in I$ s.t. $I = \langle g \rangle$, where

$$\langle g \rangle := \{gr \mid r \in R\}.$$

The element g is called a **generator** of I and I is said to be generated by g.

Example 6.29. Consider the ring $R = \mathbb{F}_2[x]/(x^3 - 1)$. The set $I := \{0, 1 + x, x + x^2, 1 + x^2\}$ is an ideal. Since I can be written as:

$$I = \{0 \cdot (1 + x), 1 \cdot (1 + x), x(1 + x), x^2(1 + x)\} = \{r(1 + x) \mid r \in R\},$$

this ideal is principal. Similarly, $J = \{rx \mid r \in R\}$ is also a principal ideal.

Definition 6.30 (Principal ideal ring)**.** A ring R is said to be a **principal ideal ring** if every ideal of R is principal.

Theorem 6.31. *The rings $\mathbb{Z}$, $\mathbb{F}_q[x]$, and $\mathbb{F}_q[x]/(x^n-1)$ are principal ideal rings.*

Theorem 6.32. *Let π be the linear map*

$$\pi : \mathbb{F}_q^n \longrightarrow \mathbb{F}_q[x]/(x^n-1), \quad u \mapsto u(x).$$

Then a nonempty subset $C \subseteq \mathbb{F}_q^n$ is a cyclic code iff $\pi(C)$ is an ideal of $\mathbb{F}_q[x]/(x^n-1)$.

Proof. ($\Leftarrow$) Suppose $\pi(C)$ is an ideal of $R = \mathbb{F}_q[x]/(x^n-1)$.

Then for $\alpha, \beta \in \mathbb{F}_q \subseteq R$ and $a, b \in C$, we have:

$$\alpha\pi(a) + \beta\pi(b) \in \pi(C).$$

Since π is linear, $\alpha\pi(a) + \beta\pi(b) = \pi(\alpha a + \beta b)$. Since π is bijective, this implies $\alpha a + \beta b \in C$. This shows C is linear.

Next, let $(c_0, \dots, c_{n-1}) \in C$. Then

$$\pi(c_0, \dots, c_{n-1}) = c_0 + c_1 x + \cdots + c_{n-1}x^{n-1} \in \pi(C).$$

Since $\pi(C)$ is an ideal, $x \cdot \pi(c) \in \pi(C)$. We compute:

$$\begin{aligned} x \cdot \pi(c) &= x \sum_{i=0}^{n-1} c_i x^i \\ &= c_0 x + c_1 x^2 + \cdots + c_{n-2}x^{n-1} + c_{n-1}x^n \\ &\equiv c_{n-1} + c_0 x + c_1 x^2 + \cdots + c_{n-2}x^{n-1} \pmod{x^n - 1}. \end{aligned}$$

Since π is bijective, we have $(c_{n-1}, c_0, \dots, c_{n-2}) \in C$.

So C is a cyclic set, and therefore a cyclic code.

($\Rightarrow$) Suppose C is cyclic.

Clearly, $\pi(C)$ is closed under "$+$" and "$-$":

$$a, b \in C \implies a \pm b \in C \implies \pi(a \pm b) \in \pi(C) \implies \pi(a) \pm \pi(b) \in \pi(C).$$

Next, let $f(x) \in \mathbb{F}_q[x]/(x^n-1)$ be given by

$$f(x) = f_0 + f_1 x + \cdots + f_{n-1}x^{n-1} = \pi(f_0, \dots, f_{n-1}).$$

Suppose further that $\pi(f_0, \dots, f_{n-1}) \in \pi(C)$. Then:

$$\begin{aligned} xf(x) &= f_{n-1} + f_0 x + \cdots + f_{n-2}x^{n-1} \pmod{x^n - 1} \\ &= \pi(f_{n-1}, f_0, \dots, f_{n-2}) \\ &\in \pi(C) \quad \text{(since } C \text{ is cyclic).} \end{aligned}$$

By induction, we have $x^i f(x) \in \pi(C)$ for all $i \geq 0$.

Thus, for any $g(x) = \sum_{i=0}^{n-1} g_i x^i \in \mathbb{F}_q[x]/(x^n - 1)$, we have:

$$\begin{aligned} g(x)f(x) &= g(x) \cdot \pi(f_0, \dots, f_{n-1}) \\ &= \sum_{i=0}^{n-1} g_i x^i \cdot \pi(f_0, \dots, f_{n-1}) \\ &= \sum_{i=0}^{n-1} \pi(g_i f_{n-i \pmod{n}}, \dots, g_i f_{n-1-i \pmod{n}}) \\ &\in \pi(C). \end{aligned}$$

Therefore, $\pi(C)$ is an ideal. □

Theorem 6.33. *Let I be a nonzero ideal in $R = \mathbb{F}_q[x]/(x^n - 1)$, i.e., $I \neq \{0\}$, and let $g(x)$ be a nonzero monic polynomial of the least degree in I. Then $I = \langle g(x) \rangle$ and $g(x) \mid x^n - 1$.*

Proof. Let $f(x) \in I$. Write $f(x) = s(x)g(x) + r(x)$, where $\deg(r(x)) < \deg(g(x))$ and $s(x), r(x) \in \mathbb{F}_q[x]$.

Since $g(x) \in I$ and $s(x) \in R$, we have $s(x)g(x) \in I$. Thus,

$$f(x) - s(x)g(x) \in I.$$

That is $r(x) \in I$. By the minimality of the degree of $g(x)$, we must have $r(x) \equiv 0$, and therefore $I = \langle g(x) \rangle$.

To show divisibility: Notice $0 \equiv x^n - 1 \pmod{x^n - 1}$. Since $0 \in I$, the polynomial $x^n - 1$ (viewed as an element in the ideal generated by $g(x)$ in $\mathbb{F}_q[x]$) implies $x^n - 1$ is divisible by $g(x)$. □

Theorem 6.34. *There is a unique monic polynomial of the least degree in every nonzero ideal I of $\mathbb{F}_q[x]/(x^n - 1)$.*

Proof. Suppose $g_1(x), g_2(x)$ are two distinct monic polynomials of the least degree in I. Then $g_1(x) - g_2(x)$ is a nonzero polynomial of smaller degree, which is a contradiction. □

Definition 6.35 (Generator polynomials)**.** The unique monic polynomial of the least degree of a nonzero ideal I of $\mathbb{F}_q[x]/(x^n - 1)$ is called the generator polynomial of I. For a cyclic code C, the generator polynomial of $\pi(C)$ is also called the generator polynomial of C.

Theorem 6.36. *Each monic divisor of $x^n - 1$ is the generator polynomial of some cyclic code in $\mathbb{F}_q^n$.*

Proof. Let $g(x)$ be monic and $g(x) \mid x^n - 1$. Define $I = \langle g(x) \rangle$. Let C be the cyclic code corresponding to I.

Assume $h(x)$ is the generator polynomial of C. Then $h(x) \in I$. Since $g(x)$ generates I, we have:

$$h(x) \equiv g(x)b(x) \pmod{x^n - 1}$$

where $b(x) \in \mathbb{F}_q[x]$. Thus, $g(x) \mid h(x)$. Since $h(x)$ has the smallest degree in I, $\deg(g(x)) \leq \deg(h(x))$. $\implies g(x) = h(x)$, i.e., $g(x)$ is the generator polynomial. $\square$

Theorem 6.37 (Dimension of cyclic codes)**.** *Let $g(x)$ be the generator polynomial of an ideal in $\mathbb{F}_q[x]/(x^n - 1)$. Then the corresponding cyclic code has dimension k if* $\deg(g(x)) = n - k$.

Proof. For $u_1(x), u_2(x)$ with $u_1(x) \not\equiv u_2(x)$ and degree at most $k - 1$, we have

$$u_1(x)g(x) \not\equiv u_2(x)g(x).$$

Consider

$$A := \{u(x)g(x) \mid u(x) \in \mathbb{F}_q[x]/(x^n - 1), \deg(u(x)) \leq k - 1\}.$$

Then $|A| = q^k$, as $u(x) = u_0 + u_1 x + \cdots + u_{k-1}x^{k-1}$ with $u_i \in \mathbb{F}_q$.

Moreover, $A \subseteq \langle g(x) \rangle$. On the other hand, for any $a(x)g(x) \in \langle g(x) \rangle$ with $a(x) \in \mathbb{F}_q[x]/(x^n - 1)$, we have:

$$a(x)g(x) = u(x)(x^n - 1) + v(x)$$

where $\deg(v(x)) < n$. Since $g(x) \mid a(x)g(x)$ and $g(x) \mid (x^n - 1)$, it follows that:

$$g(x) \mid (a(x)g(x) - u(x)(x^n - 1)) = v(x).$$

So we can write $v(x) = g(x)b(x)$ for some $b(x)$. Therefore,

$$\deg(b(x)) = \deg(v(x)) - \deg(g(x)) < n - \deg(g(x)).$$

Since $\deg(g(x)) = n - k$, we have $\deg(b(x)) < k$. Because $v(x) = b(x)g(x)$ and $\deg(b(x)) < k$, we have $v(x) \in A$.

Hence, $A = \langle g(x) \rangle$, and

$$\dim_{\mathbb{F}_q} \langle g(x) \rangle = \log_q |A| = k.$$

$\square$

Theorem 6.38 (Generator matrices)**.** *Let $g(x) = g_0 + g_1 x + \cdots + g_{n-k}x^{n-k}$ be the generator* *polynomial of a cyclic code C in $\mathbb{F}_q^n$ with $\deg(g(x)) = n-k$ (i.e., $g_{n-k} \neq 0$). Then the matrix*

$$G = \begin{pmatrix} g_0 & g_1 & \dots & g_{n-k} & 0 & \dots & 0 \\ 0 & g_0 & g_1 & \dots & g_{n-k} & \dots & 0 \\ \vdots & & \ddots & \ddots & & \ddots & \vdots \\ 0 & 0 & \dots & g_0 & g_1 & \dots & g_{n-k} \end{pmatrix}$$

consisting of k rows, is a generator matrix for C.

Proof. By definition, $C = \{u(x)g(x) \mid u(x) \in \mathbb{F}_q[x]/(x^n-1), \deg(u(x)) \leq k-1\}$.

The polynomial $g(x)$ corresponds to the first row of G:

$$g(x) \xrightarrow{\pi^{-1}} (g_0, g_1, \dots, g_{n-k}, \underbrace{0, \dots, 0}_{k-1})$$

The subsequent rows are cyclic shifts, which correspond to multiplying by x^i:

$$(\underbrace{0, \dots, 0}_{i}, g_0, \dots, g_{n-k}, \underbrace{0, \dots, 0}_{k-1-i}) \longleftrightarrow x^i g(x), \quad i \leq k-1.$$

The set of rows is $\{g(x), xg(x), \dots, x^{k-1}g(x)\}$. Any codeword can be expressed as:

$$\begin{aligned} u(x)g(x) &= g(x)\sum_{i=0}^{k-1} u_i x^i \\ &= u_0(g(x)) + u_1(xg(x)) + \dots + u_{k-1}(x^{k-1}g(x)) \end{aligned}$$

So G generates C. □

What about parity-check matrices for C?

Generator matrices of $C^\perp$ are parity-check matrices for C.

$$\boxed{C \text{ is cyclic } \iff C^\perp \text{ is cyclic.}}$$

Definition 6.39. Let $g(x)$ be a generator polynomial of a cyclic code in $\mathbb{F}_q^n$. Define $h(x) = \frac{x^n-1}{g(x)}$. $h(x)$ is called the *check polynomial* of C.

Let $c(x) = f(x)g(x)$ be a codeword of C. Then $c(x)h(x) = f(x)g(x)h(x) \equiv 0 \pmod{x^n - 1}$. Define $k = \deg(h(x))$. Consider the coefficient of x^j in $c(x)h(x)$:

$$c(x)h(x) = \left(\sum_{i=0}^{n-1} c_i x^i\right)\left(\sum_{t=0}^{k} h_t x^t\right)$$

For $0 \leq j \leq n-1$:

$$0 = \sum_{i=0}^{n-1} c_i h_{j-i} \text{ is the coefficient of } x^j,$$

where the subscript of h_{j-i} is taken modulo n ($j - i \equiv n + j - i \pmod{n}$).

Note that $x^j \equiv x^{n+j} \pmod{x^n - 1}$.

$$\underbrace{\begin{pmatrix} 0 & \dots & 0 & h_k & \dots & h_0 \\ 0 & \dots & h_k & \dots & h_0 & 0 \\ \vdots & & & & & \vdots \\ h_k & \dots & h_0 & \dots & \dots & 0 \end{pmatrix}}_{H} \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{pmatrix} = 0$$